

Introduction of OpenF2A

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A photograph of a wind farm at sunset. The sky is a mix of orange, yellow, and blue. Several wind turbines are visible, their silhouettes reflected in a body of water in the foreground. The image is partially obscured by a white diagonal shape on the right side.

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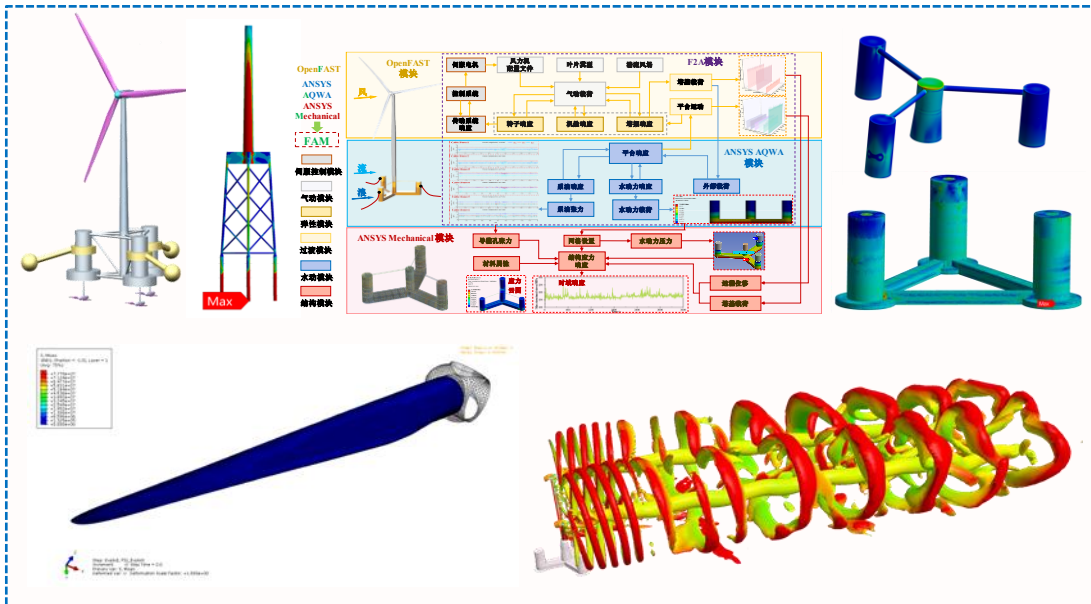
User guide

CV of Yang Yang



Yang Yang

- 2009.09-2020.04, BsC, PhD, University of Shanghai for Science and Technology, P.R.China
- 2017.10-2018.10, Visiting PhD funded by CSC, Liverpool John Moores University, UK
- 2018.10-2020.11, Post-doctoral Research Associate, Liverpool John Moores University, UK
- 2021.05-2024.12, Associate Professor, Ningbo University, P.R. China
- 2024.12-2025.09, Professor, Ningbo University, P.R. China

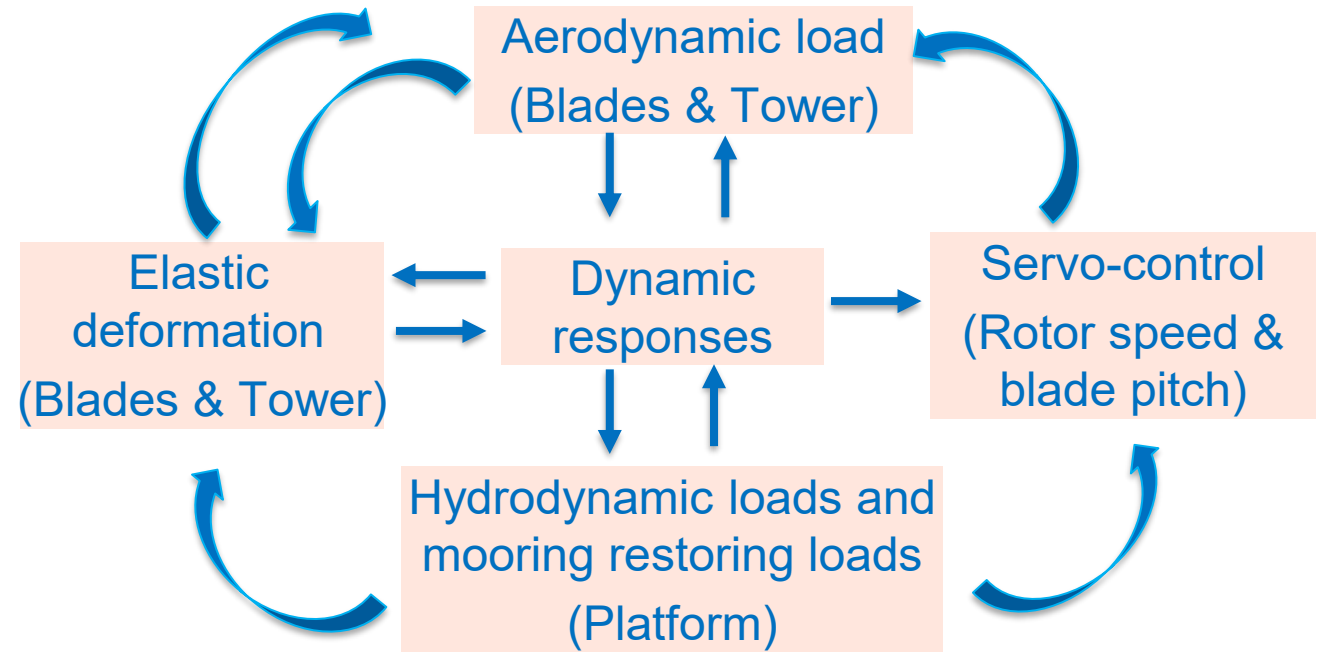
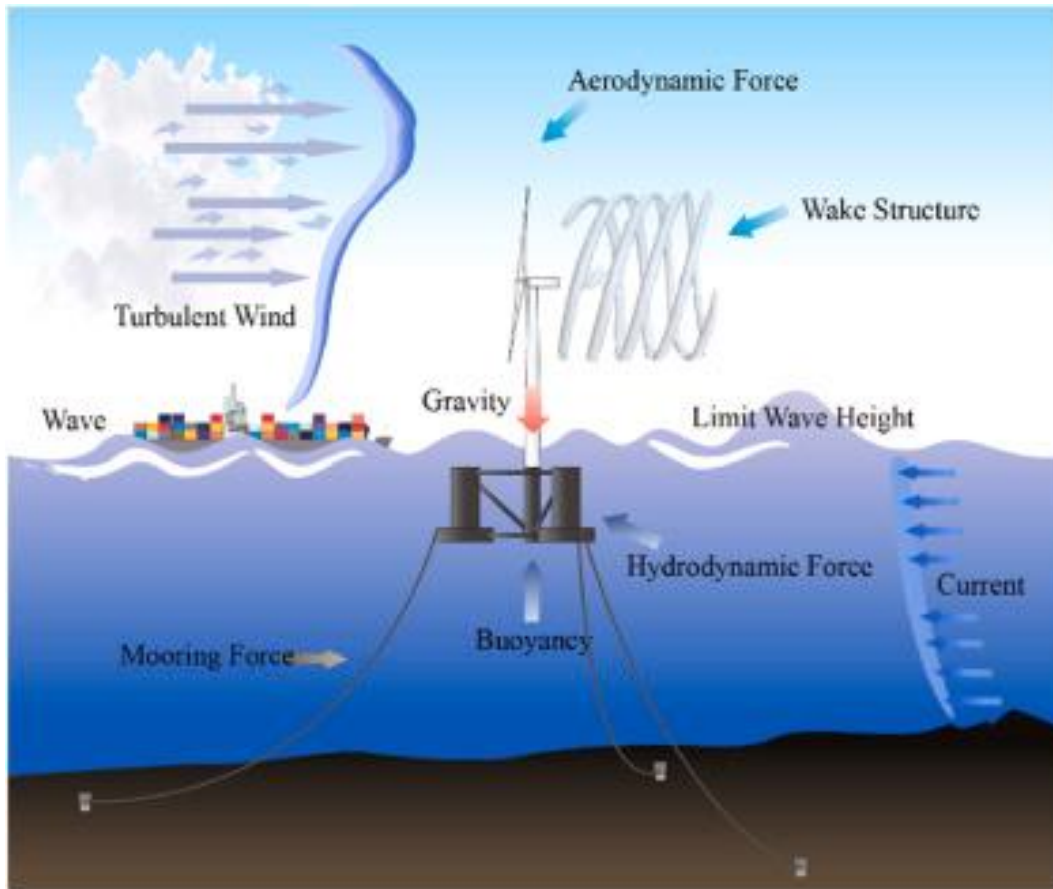


Research Interests

- Numerical modelling of wind-wave-current integrated floating energy systems
- Numerical modelling of twin-rotor floating offshore wind turbines
- Structural design of floating offshore wind platforms
- Aeroelastic modelling of offshore wind turbine blades
- Seismic analysis and structural control

<https://scholar.google.com/citations?user=frTAwNQAAAAJ&hl=zh-CN>

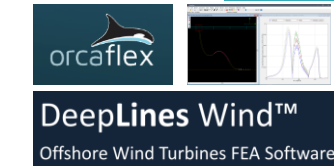
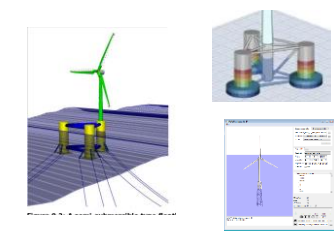
Fully coupled effects of FOWTs



**Aero-Hydro-Servo-Elastic
Coupling of Floating OWTs**

Main numerical tools

Tool	Developer	Notes
Bladed	DNV in Norway	Earliest commercial software for wind turbine dynamic analysis,
SESAM	DNV/Sintef in Norway	Almost the only option for floating wind platform structural design
HAWC2	DTU in Denmark	One of the earliest tool, FEM for structures, without GUI, reliable aero-elastic
OrcaFlex	Orcina in UK	Reliable simulation tool for cables, and mooring lines, getting stronger
DeepLines Wind	PRINCIPIA in France	Suitable for VAWT/HAWT, FEM for structural analysis
Qblade	BTU in Germany	Open-source, FVW involved for aerodynamics, powerful for different FOWTs
Simpack	SIMULIA in France	Coupled with many OpenFAST modules, capable of modelling fractured blades
Ashes	Simis in Norway	Developed by graduates from NTNU, with good GUI
3DFloat	IFE in Norway	FEM for structural analysis, excellent in pile-soil interaction
OpenFAST	NREL in US	Most popular open-source numerical tool, widely used in academic area, still being developed
Samcef	Samtech in Belgium	Developed by SAMTECH, capable of modeling nonlinear pile-soil interaction



Main numerical tools

MI&T TimeFloat-FAST

Marine Innovation
& Technology

Developed by MI&T by integrating FAST into hydrodynamic analysis tool TimeFloat. Dynamic response of the wind turbine are solved in FAST, responses of the mooring system and floating platform are predicted in TimeFloat.

Used for the design of WindFloat concept

Texas A&M CHARM3D-FAST

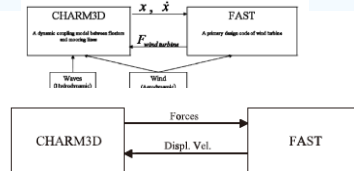


Fig. 3. Basic concept of CHARM3D-FAST hybrid model.

Similar to TimeFloat-FAST, this tool consists of a hydrodynamic analysis tool and FAST. It was originally developed by a master student in Texas A&M University in 2007 based on FAST v6. Displacement and velocity from CHARM3D are transferred to FAST and aerodynamics from FAST are fed back. Acceleration of the platform is ignored. However, the coupling relationship between FAST and CHARM3D in its later application is in the contrast to that was implemented previously.

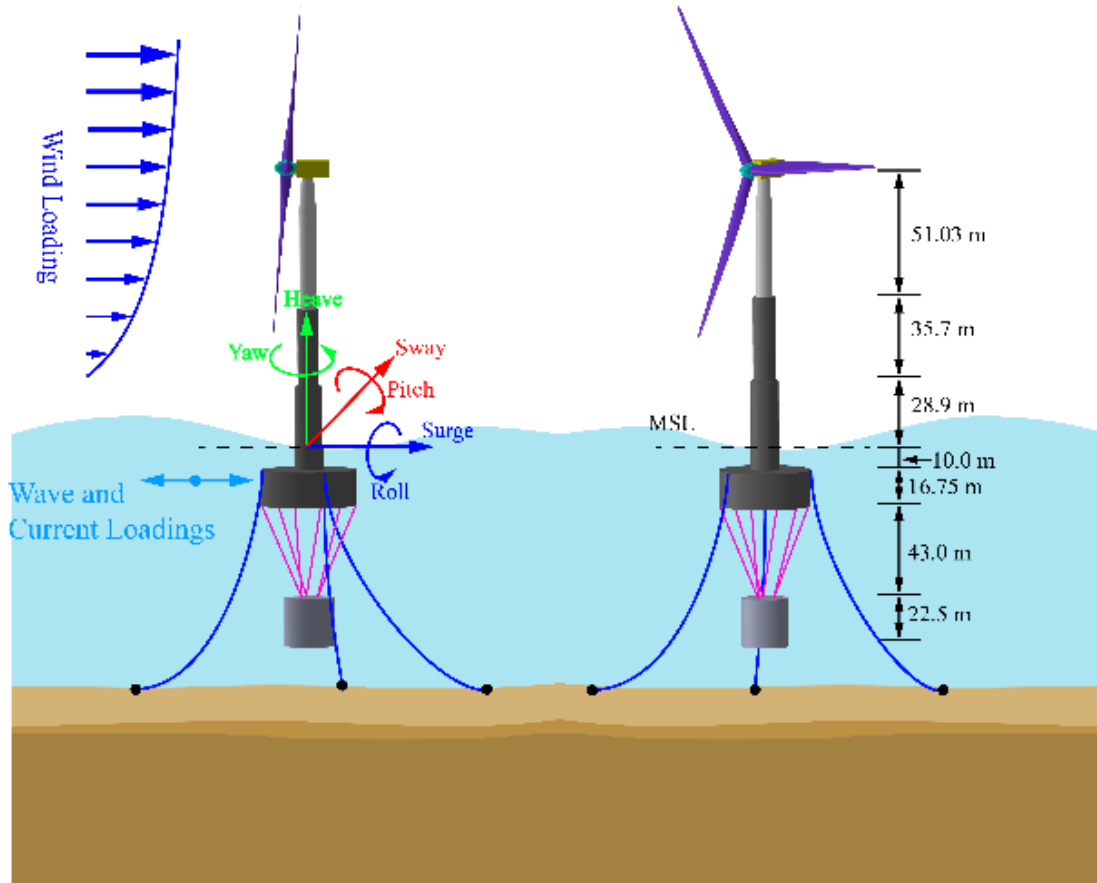
Yang YANG FAST2AQWA(F2A)



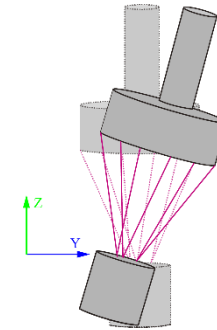
F2A is developed by Dr Yang Yang in Liverpool John Moores University for fully coupled analysis of a multi-body floating wind platform. Now it is released to the public on GitHub: <https://github.com/yang7857854/F2A>.

Similar to TimeFloat, dynamic responses of the wind turbine are predicted in OpenFAST and the responses of the platform and mooring lines are solved in AQWA.

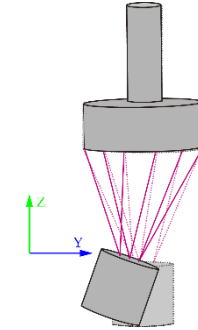
Motivation of development of F2A



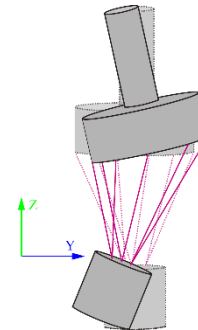
Telwind concept



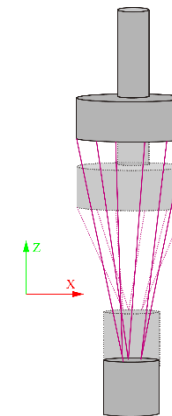
Collective pitch
36.5 s



Local pitch
3.08 s

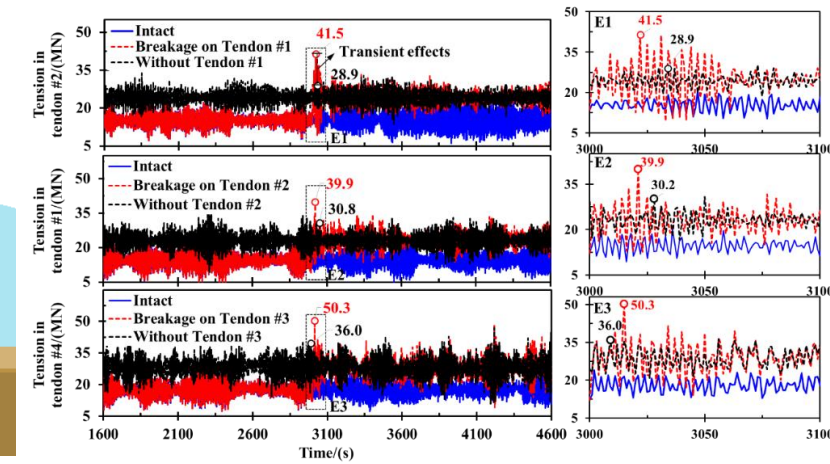
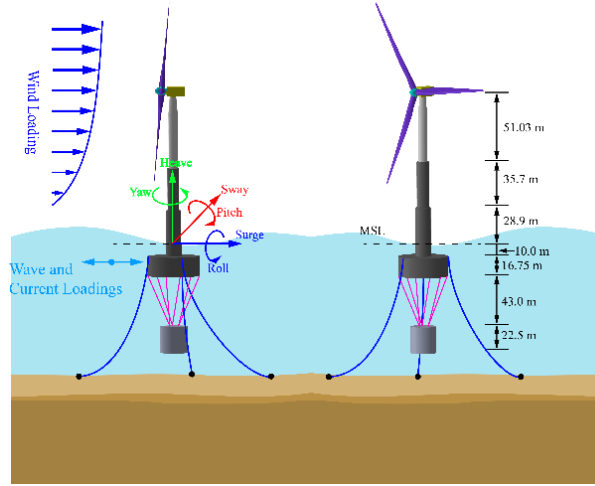
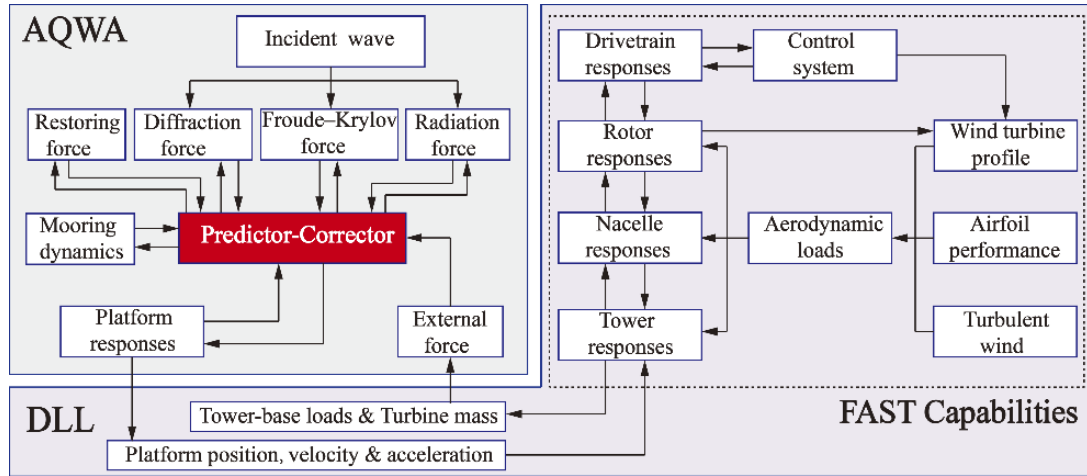


Coupling pitch
1.15 s



Relative heave
0.71 s

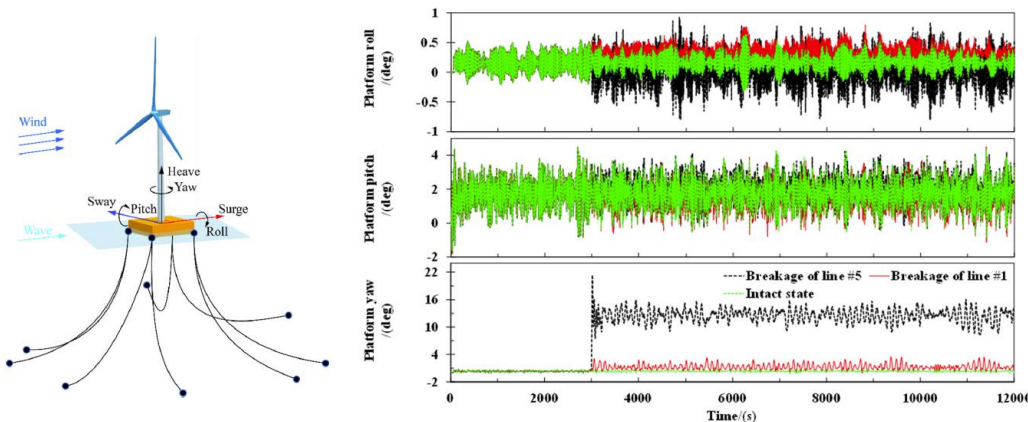
Development of F2A and its limitation



Framework of F2A (Yang Y, Bashir M, Michailides C, et al. Renewable Energy, 2020, 161: 606-625.)

Application for the Telwind concept

Yang Y, Bashir M, Michailides C, et al. Renewable Energy, 2021, 176, 89-105



Application for mooring breakage analysis

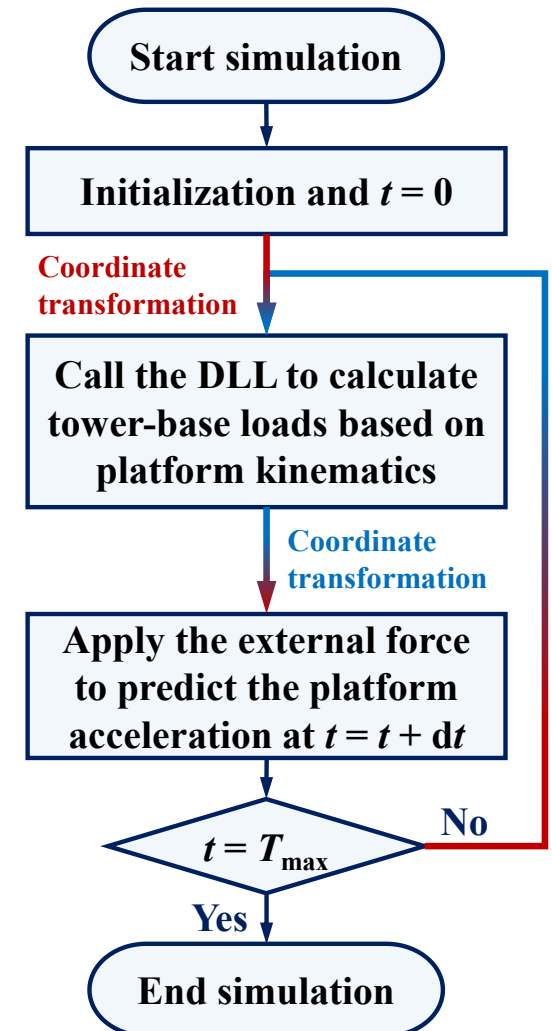
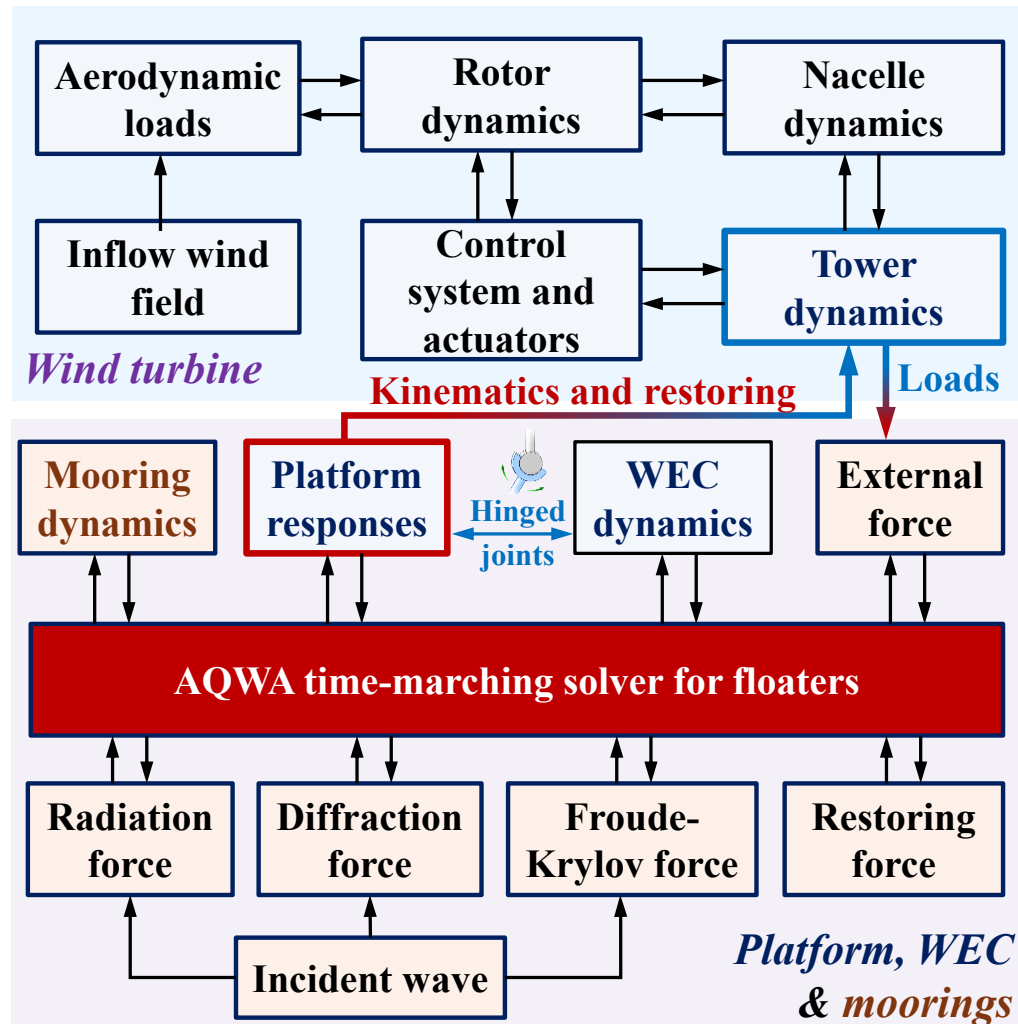
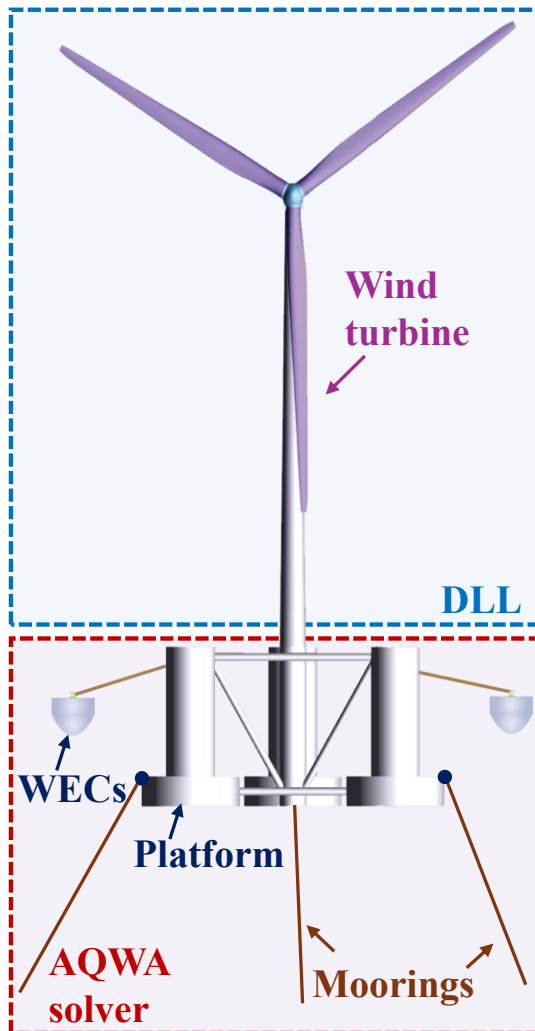
Yang Y, Bashir M, Li C, et al. Ocean Engineering, 2021, 233: 108887.

- ✓ F2A cannot accurately capture the **nonlinear aeroelastic** behavior of long, flexible blades used in large-scale FOWTs
- ✓ Fails for **eccentric** FOWT concepts, e.g WindFloat
- ✓ Underestimation of the **tower's natural frequency** without consideration of the hydrostatic stiffness and added mass of the platform



Development

Framework of OpenF2A



Logic diagram of the OpenF2A framework

Transformation of platform motions

Platform motions **output** by the Solver
(Displacement, Velocity, Acceleration)

Platform motions **transmit** to the DLL
(Displacement, Velocity, Acceleration)

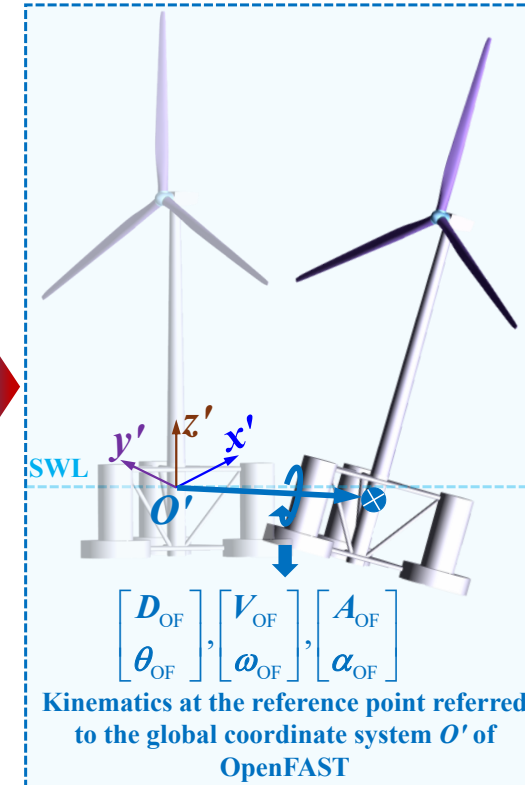
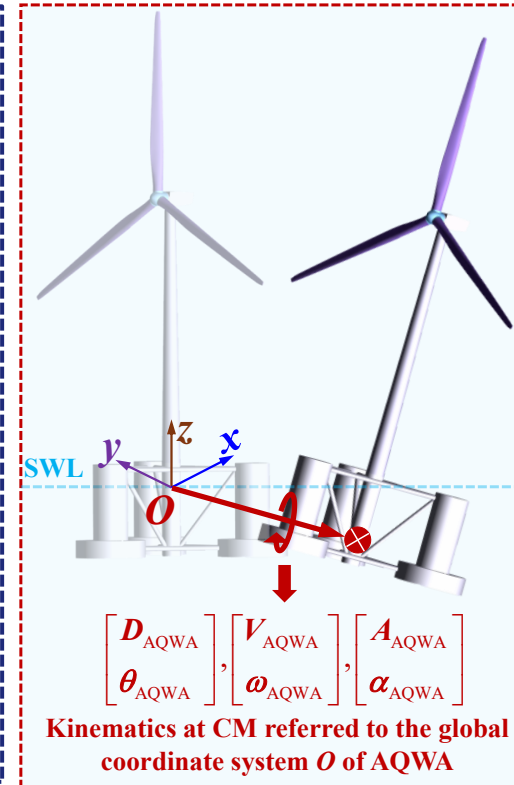
Rigid body motion
transformation
based on Euler Angle

Translational displacement : $D_{OF} = D_{AQWA} - T_{Euler} \cdot L_{ref} - P_{OO'}$

Translational velocity: $V_{OF} = V_{AQWA} - T_{Euler} \cdot L_{ref} \times \omega_{AQWA}$

Acceleration: $A_{OF} = (V_{OF} - V'_{OF})/dt$

$$T_{Euler} = \begin{bmatrix} cy \cdot cz & sx \cdot sy \cdot cz - cx \cdot sz & cx \cdot sy \cdot cz + sx \cdot sz \\ cy \cdot sz & sx \cdot sy \cdot sz + cx \cdot cz & cx \cdot sy \cdot sz - sx \cdot cz \\ -sy & sx \cdot cy & cx \cdot cy \end{bmatrix}$$



Transformation of tower-base loads

Loads **output** by the DLL
(Force and moment)

Tower-base or **local coordinate system**

Rigid body motion transformation
based on Euler Angle

Translational loads: $F = T_{\text{Euler}} \cdot F_p$

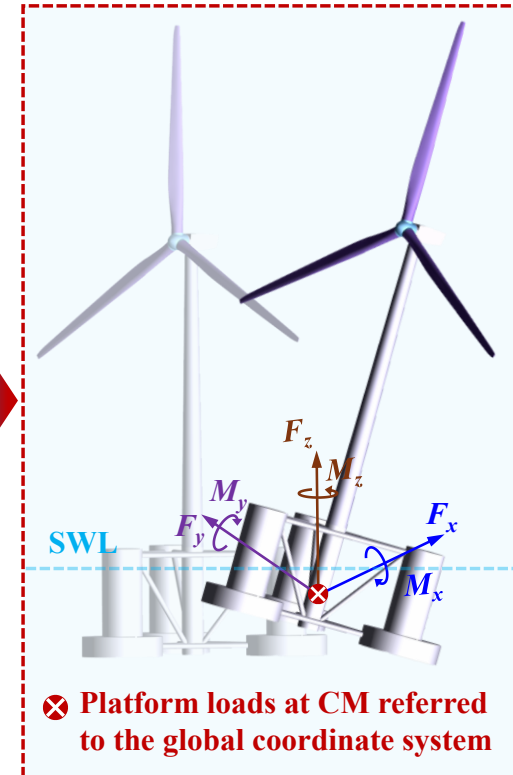
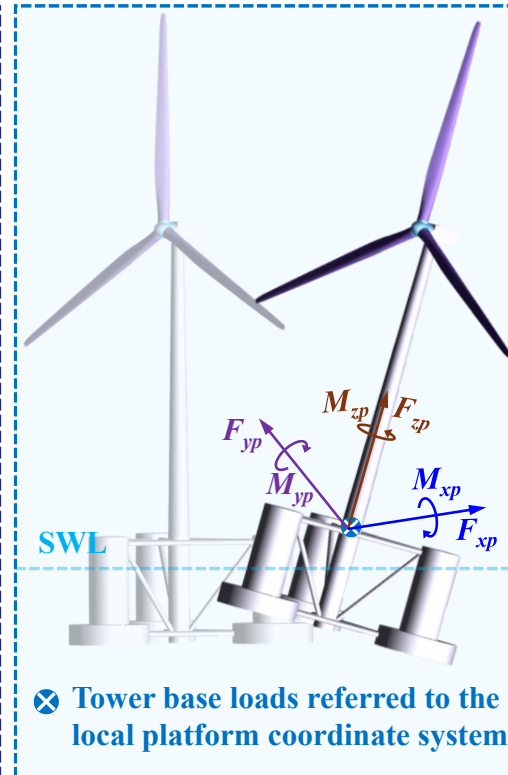
Rotating loads: $M = T_{\text{Euler}}^{-1} \cdot [M_p + (L_{\text{ref}} + P_{OO'}) \times F_p]$

Tower-base $\rightarrow$ **CM**

Local coordinate system $\rightarrow$ **Global coordinate system**

Loads **transmit** to the Solver
(Force and moment)

CM or **global coordinate system**

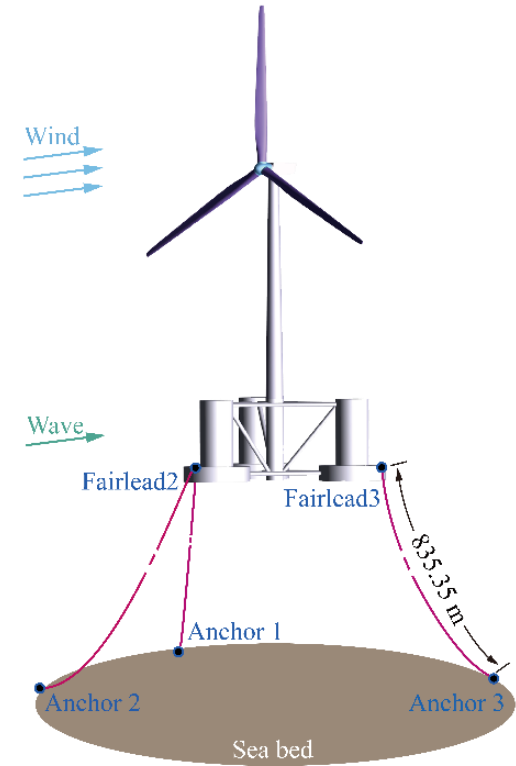
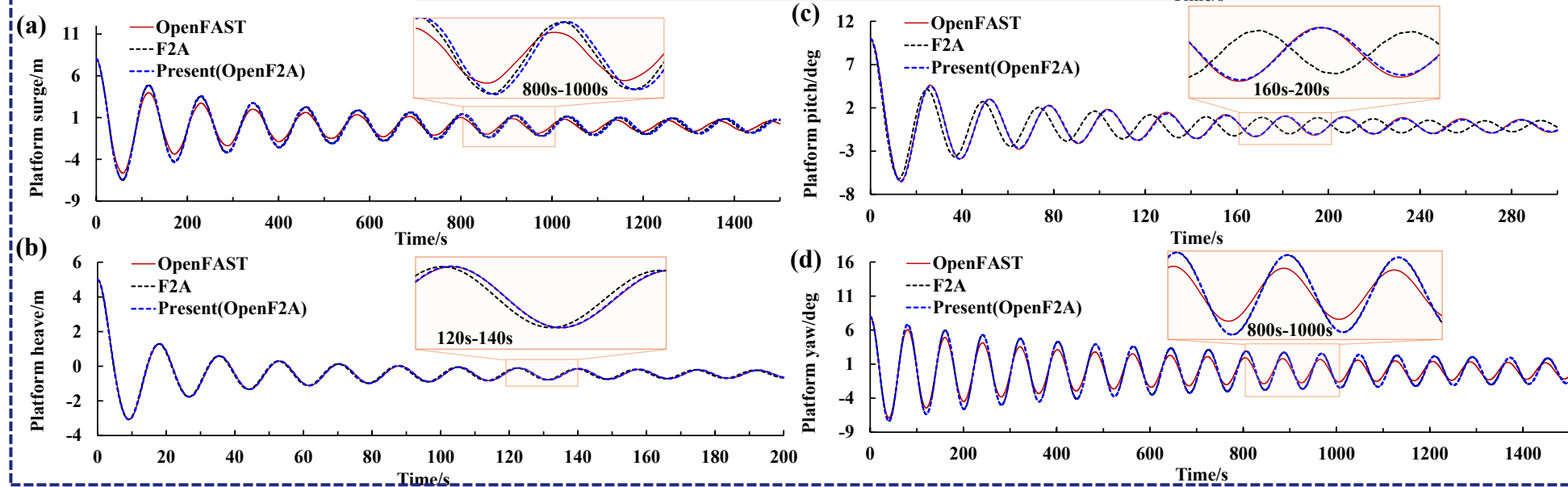




Model validations

5MW DeepCWind model

Free decay analysis of platform motions



DeepCWind 5MW wind turbine model

Comparisons of natural periods of the FOWT (Unit: s)

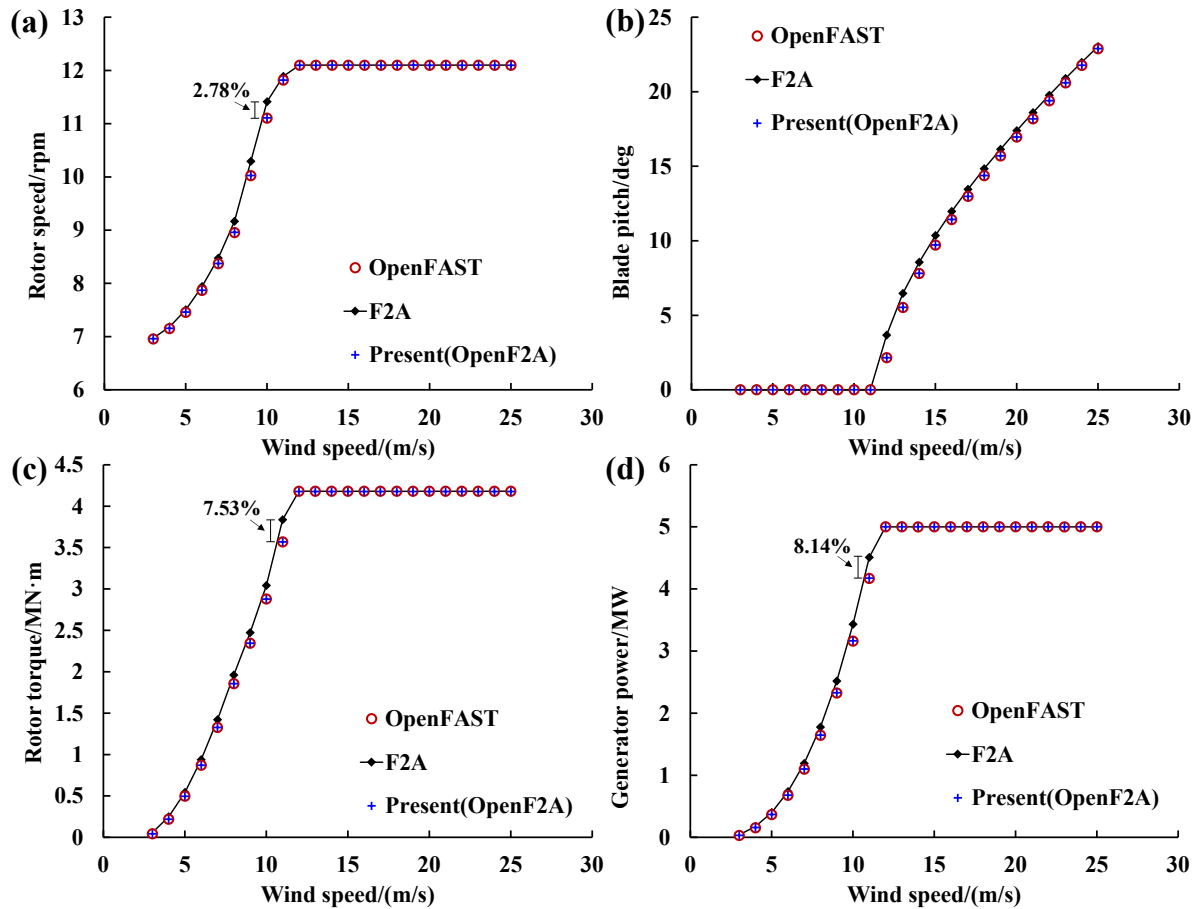
	OpenFAST	F2A	OpenF2A
Surge	114.14	114.46	115.01
Heave	17.56	17.46	17.54
Pitch	25.86	24.47	25.87
Yaw	80.45	80.62	80.67

The results obtained using OpenF2A framework agree well with the predictions of OpenFAST.

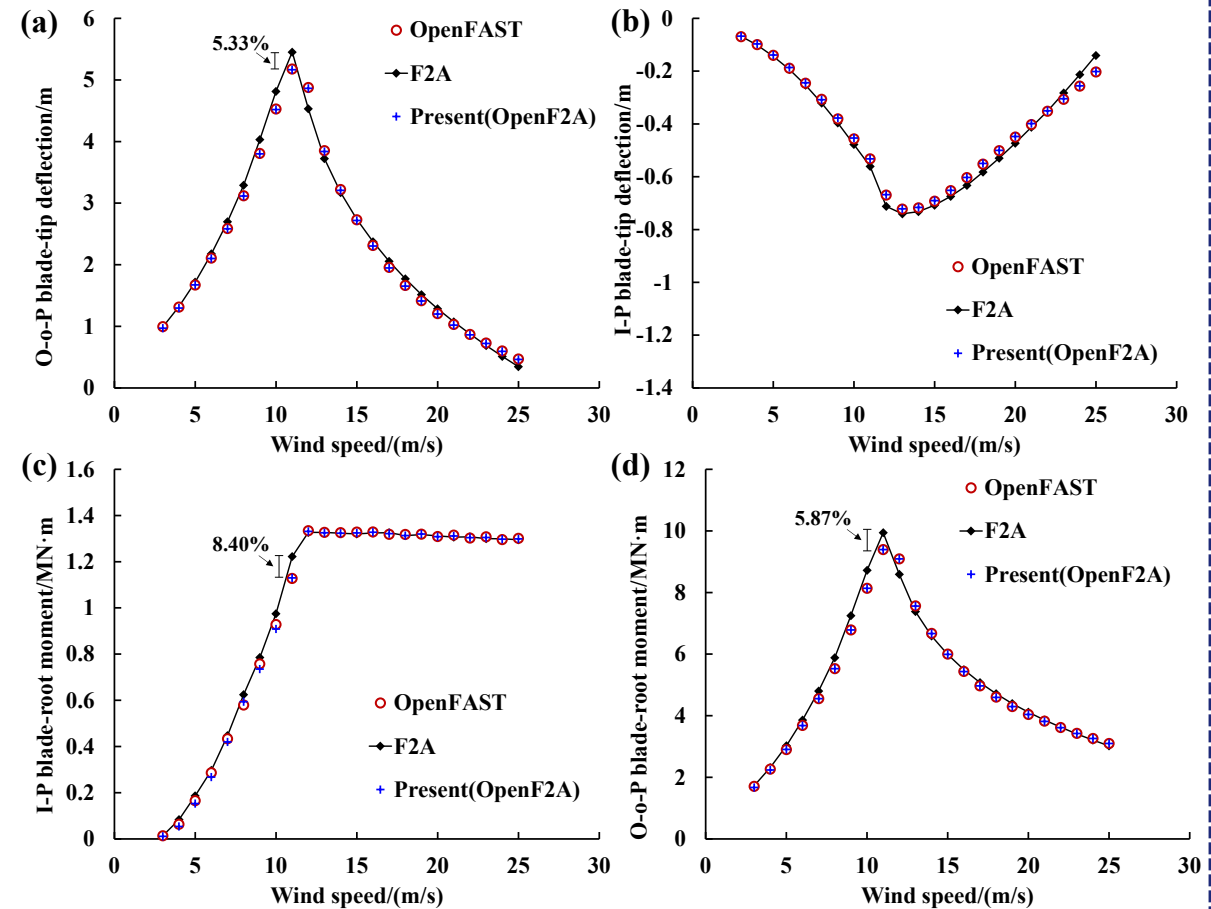
OpenF2A is improved by retaining the terms associated with platform DOFs in the **wind turbine's equation of motion**, compared to previous F2A.

5MW DeepCWind model

Drivetrain and control system

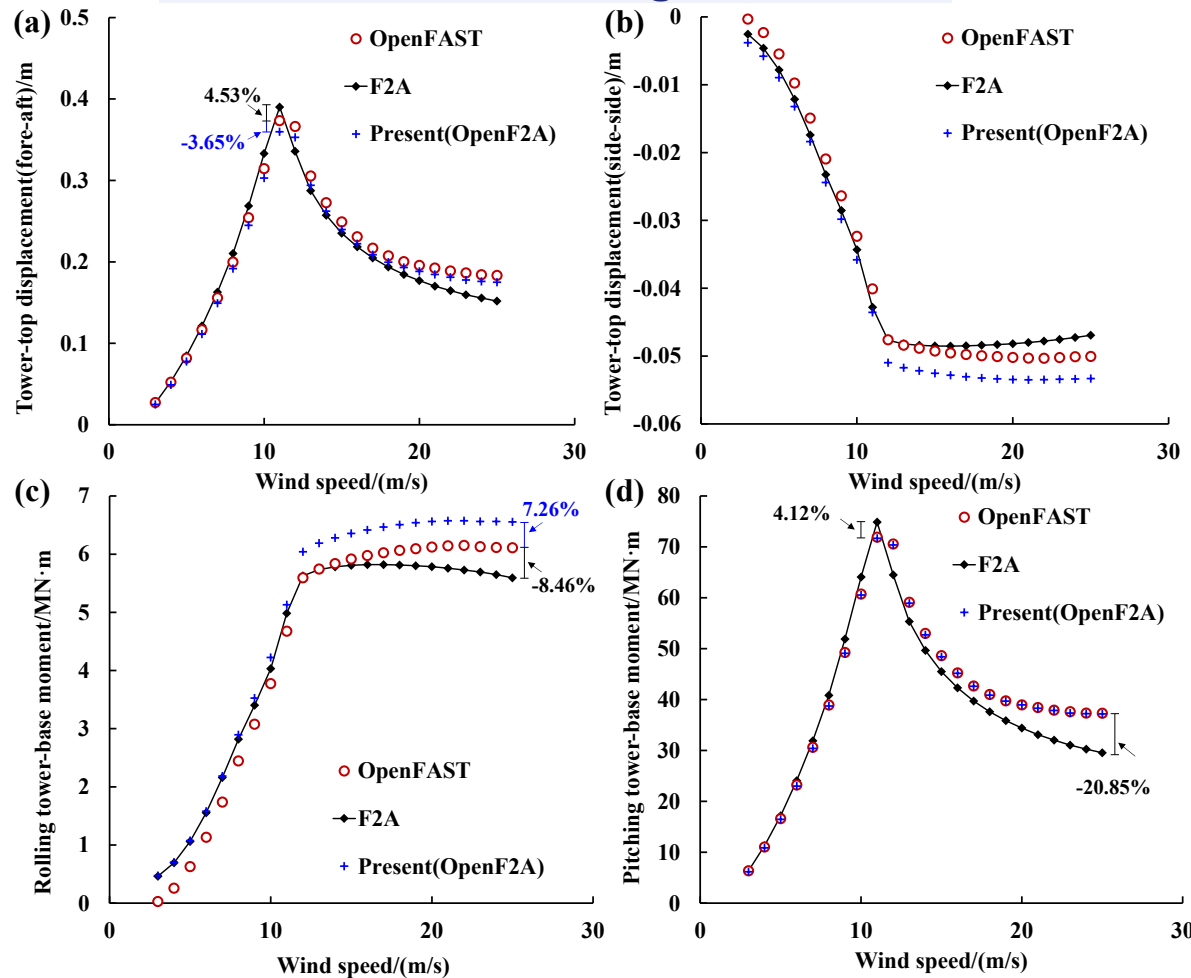


Blade-tip deflections/blade-root bending moments

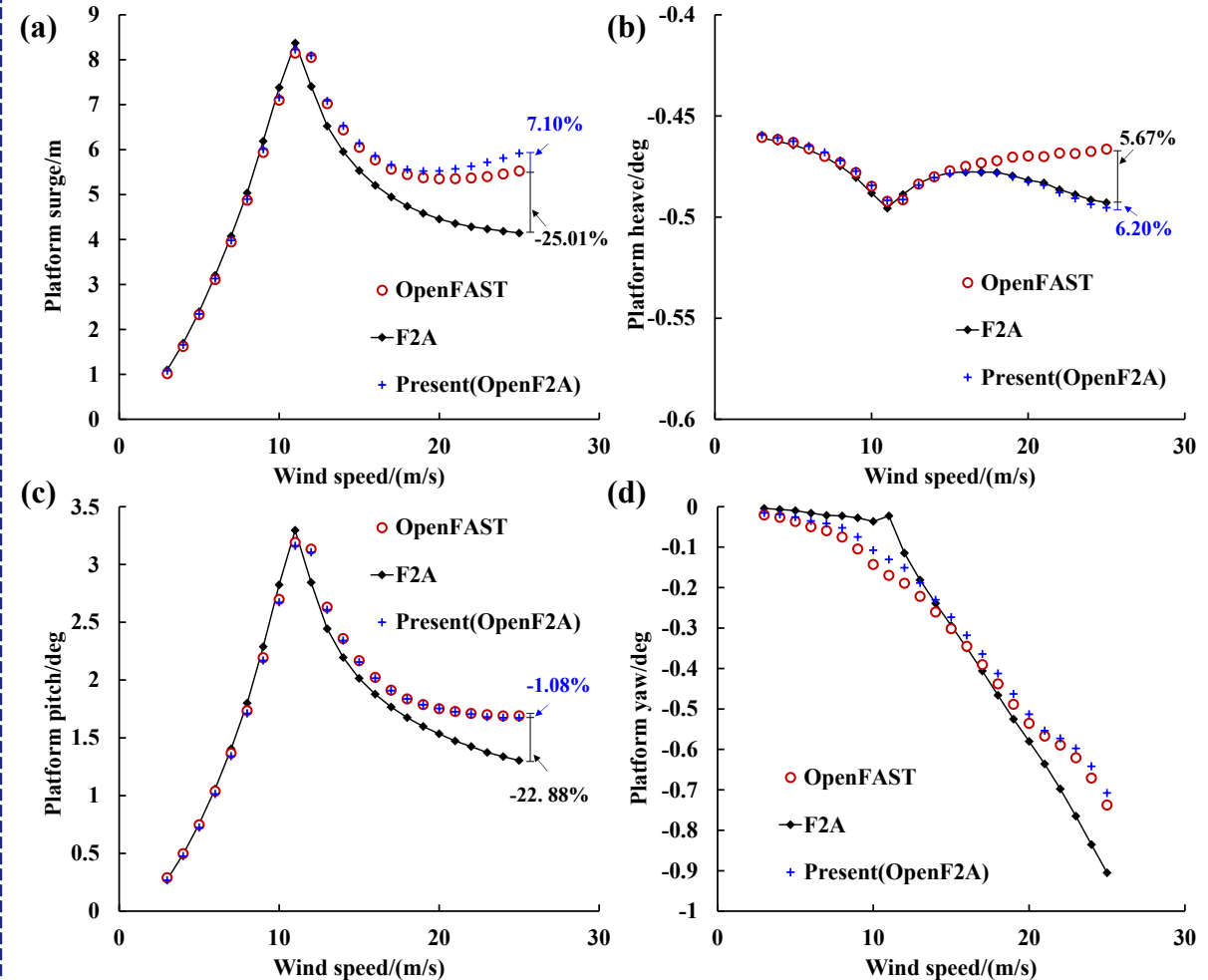


5MW DeepCWind model

Tower-top displacements/ Tower-base bending moments

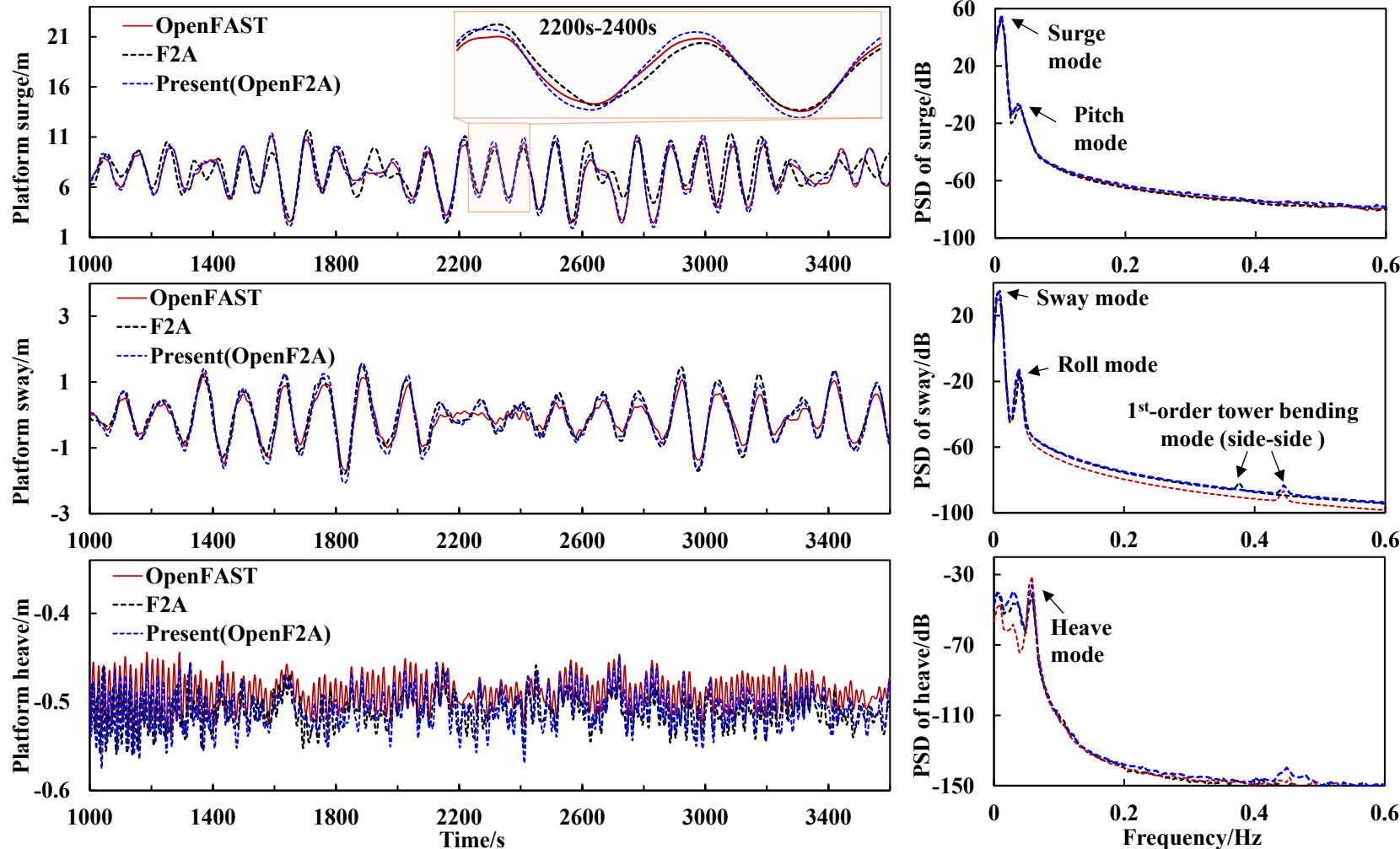


Platform motions



5MW DeepCWind model

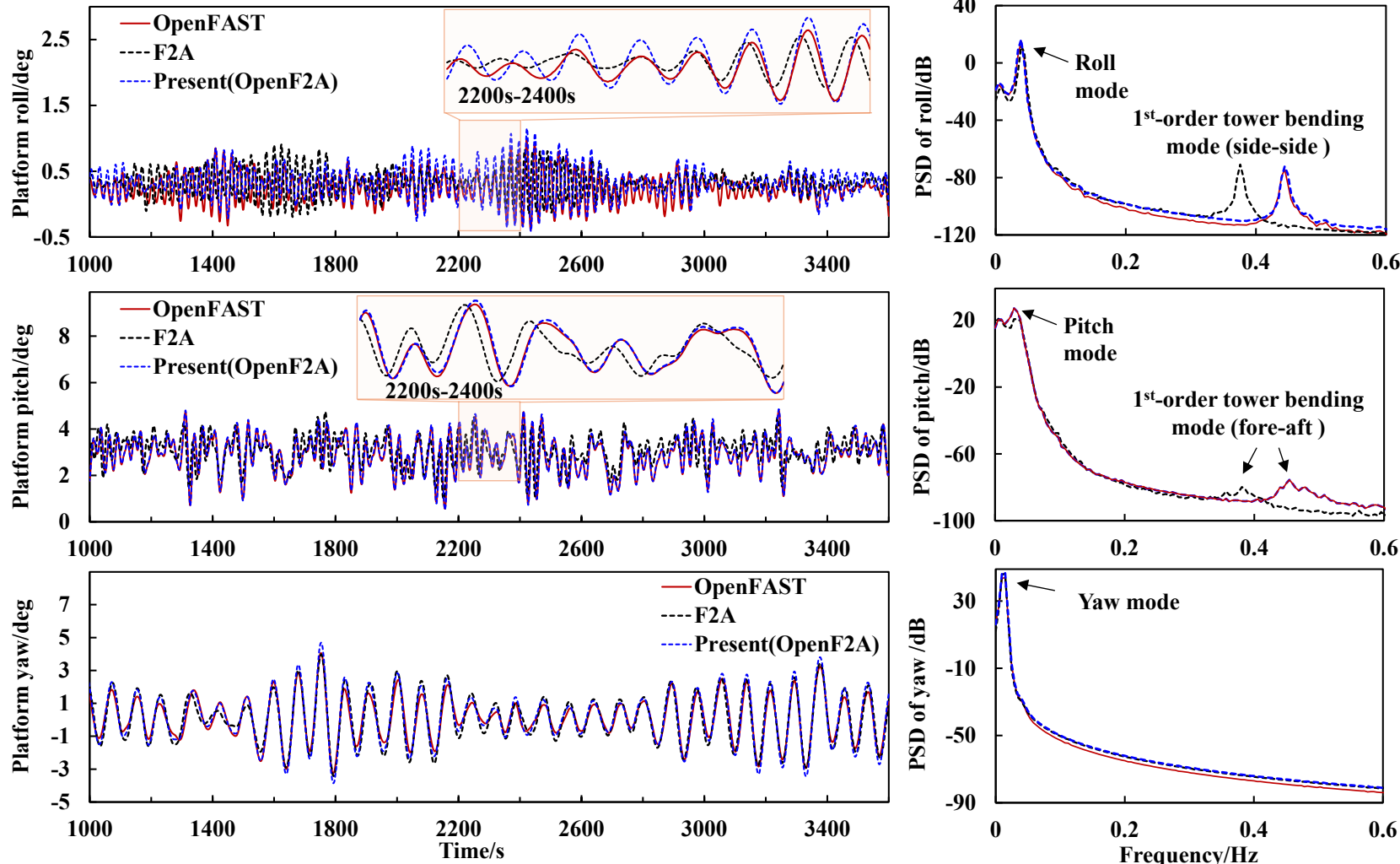
Translational platform motions and the corresponding PSD distribution



- The variation trend of platform surge and sway predicted by OpenF2A aligns with that calculated by OpenFAST.
- The comparison of PSD distributions indicates that discrepancies of heave motion are mainly attributed to the surge and heave modes.

5MW DeepCWind model

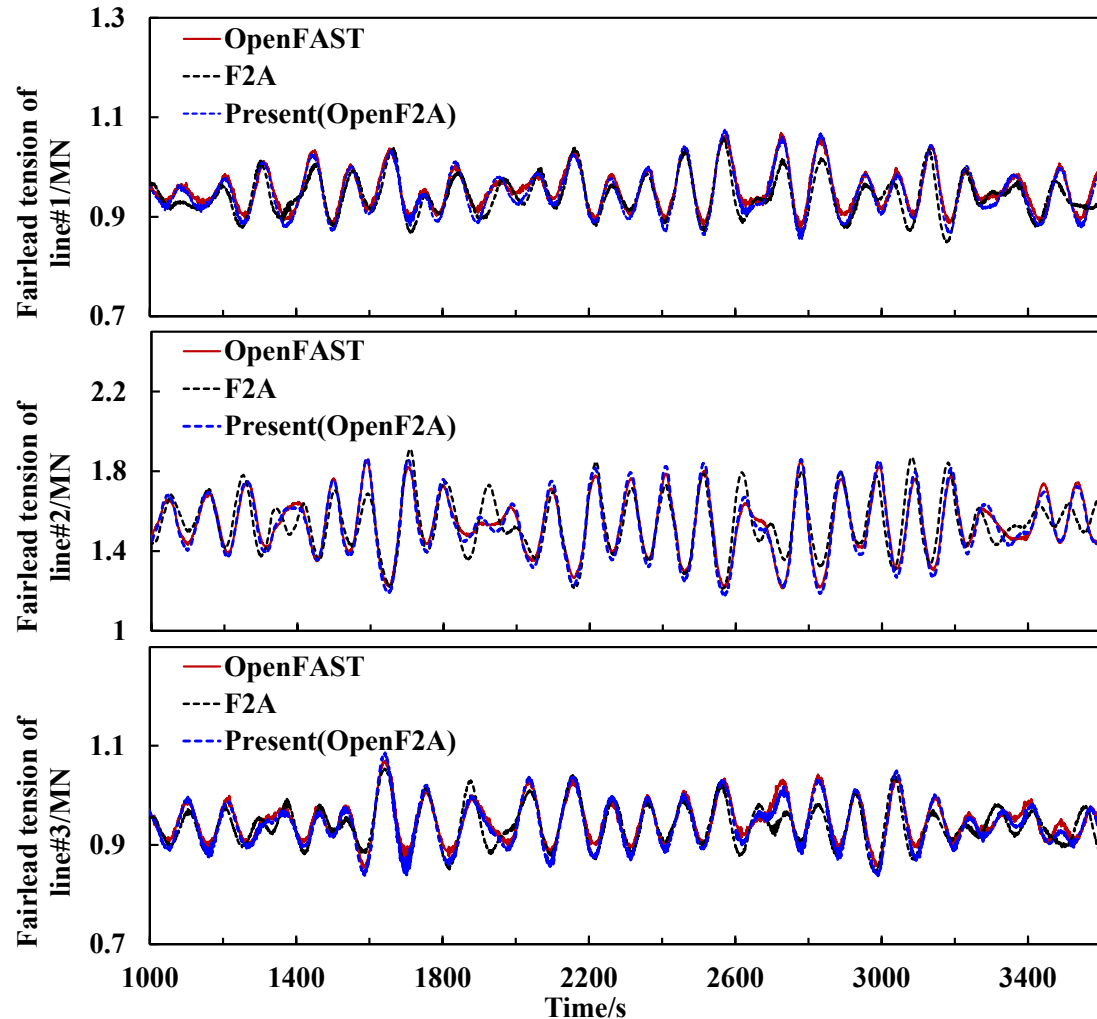
Angular platform motions and the corresponding PSD distribution



- Good agreements are obtained between the pitch motions calculated by OpenF2A and OpenFAST, although slight difference in time variations is observed.
- OpenF2A is capable of producing reliable platform responses of FOWTs.

5MW DeepCWind model

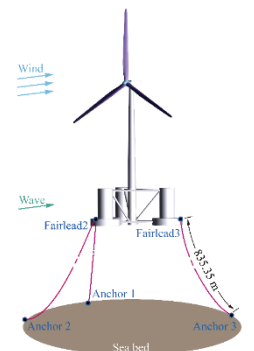
Fairlead tension in mooring lines



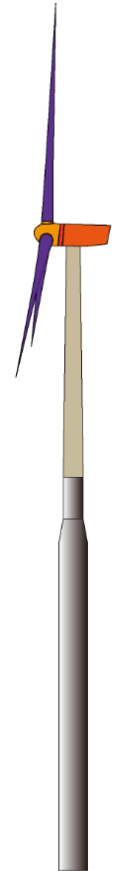
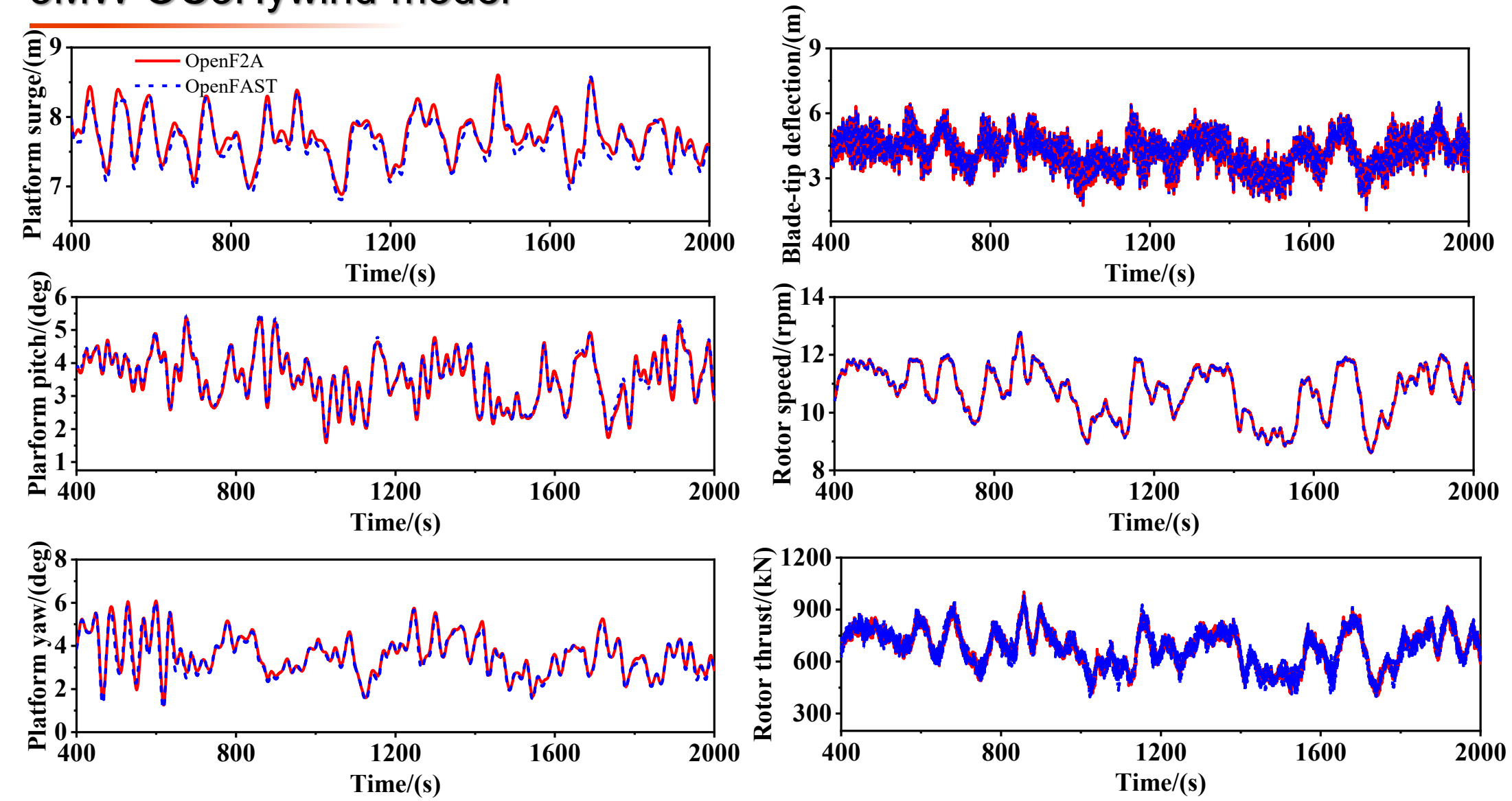
Statistical tension in the mooring lines

		OpenFAST	F2A		Present (OpenF2A)	
		Value (MN)	Value (MN)	Difference	Value (MN)	Difference
Line#1	Mean	0.955	0.942	1.34%	0.947	0.75%
	Min	0.874	0.849	2.94%	0.853	2.41%
	Max	1.070	1.060	0.98%	1.077	-0.57%
Line#2	Mean	1.537	1.541	-0.32%	1.530	0.46%
	Min	1.213	1.212	0.06%	1.177	2.92%
	Max	1.855	1.911	-3.04%	1.869	-0.78%
Line#3	Mean	0.949	0.940	0.96%	0.942	0.70%
	Min	0.851	0.844	0.87%	0.835	1.86%
	Max	1.073	1.054	1.77%	1.087	-1.30%

The discrepancies of mooring tension predicted by OpenF2A compared to OpenFAST and F2A are **less than 4%**.

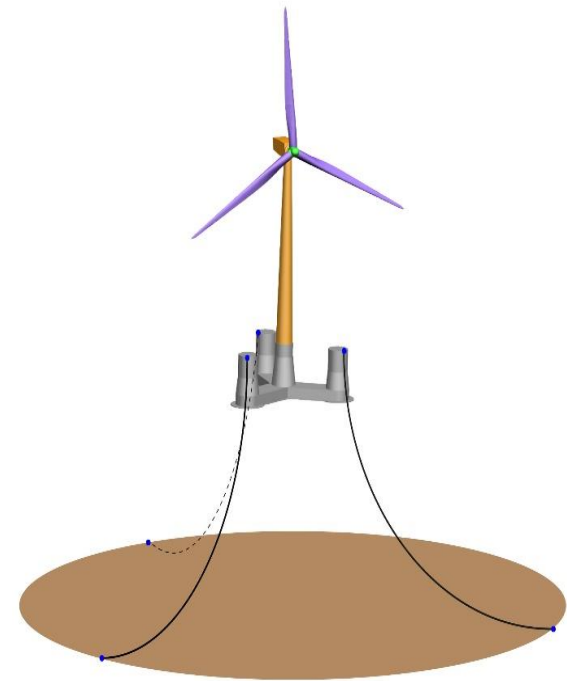
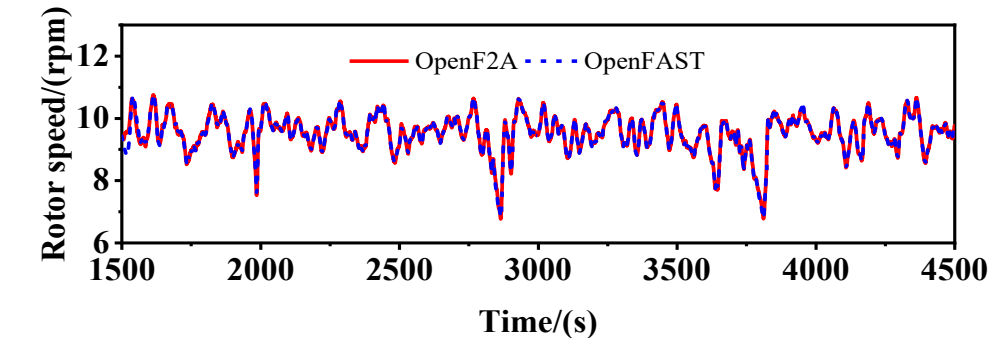
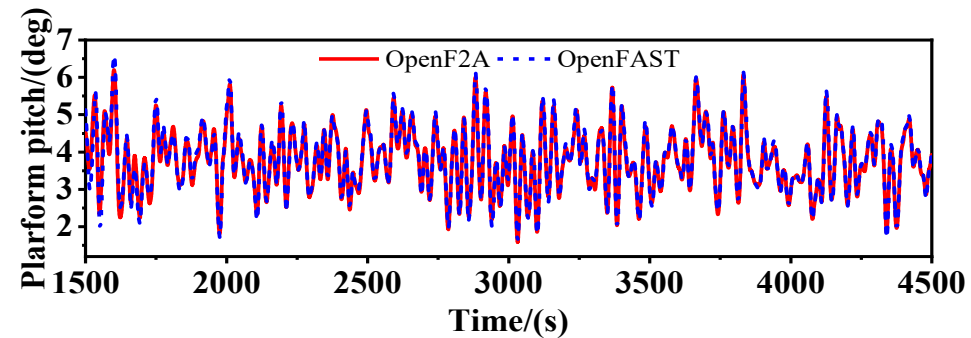
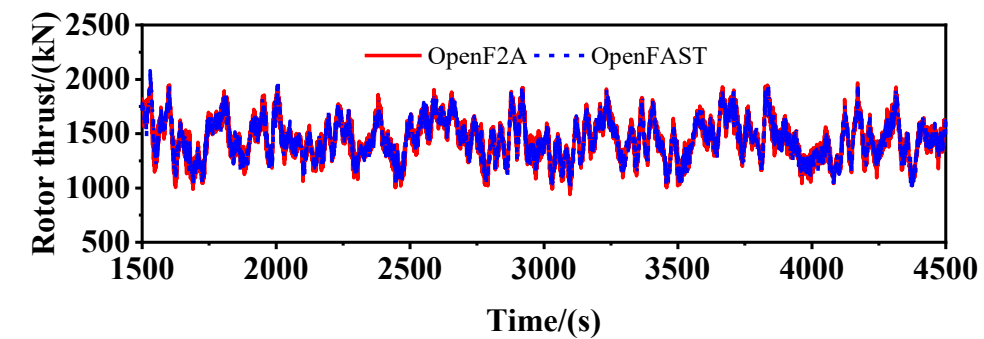
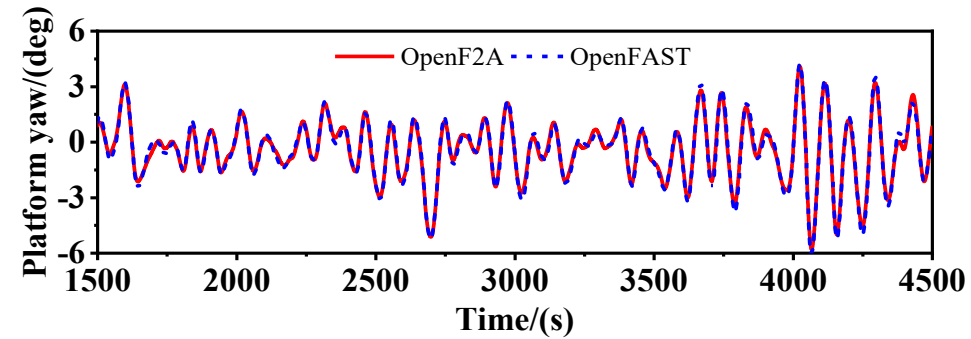
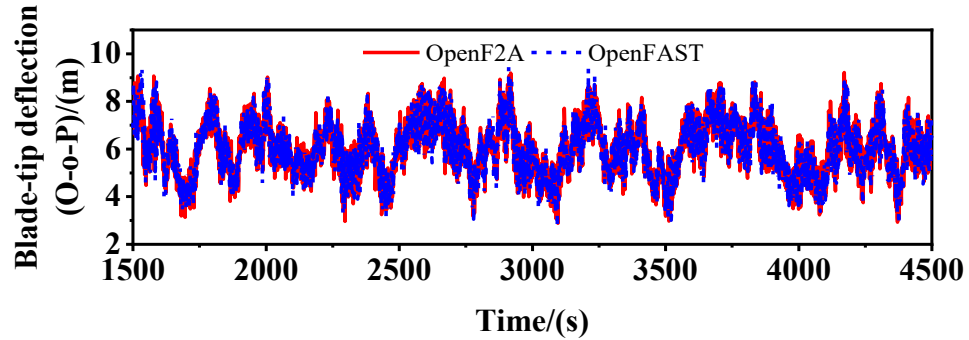
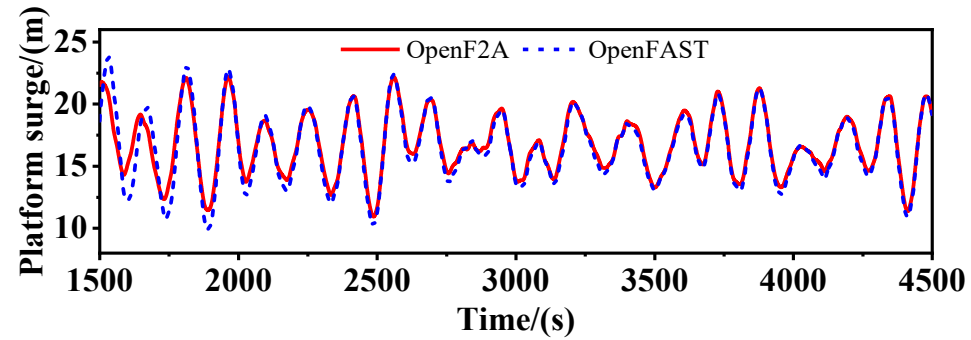


5MW OC3Hywind model



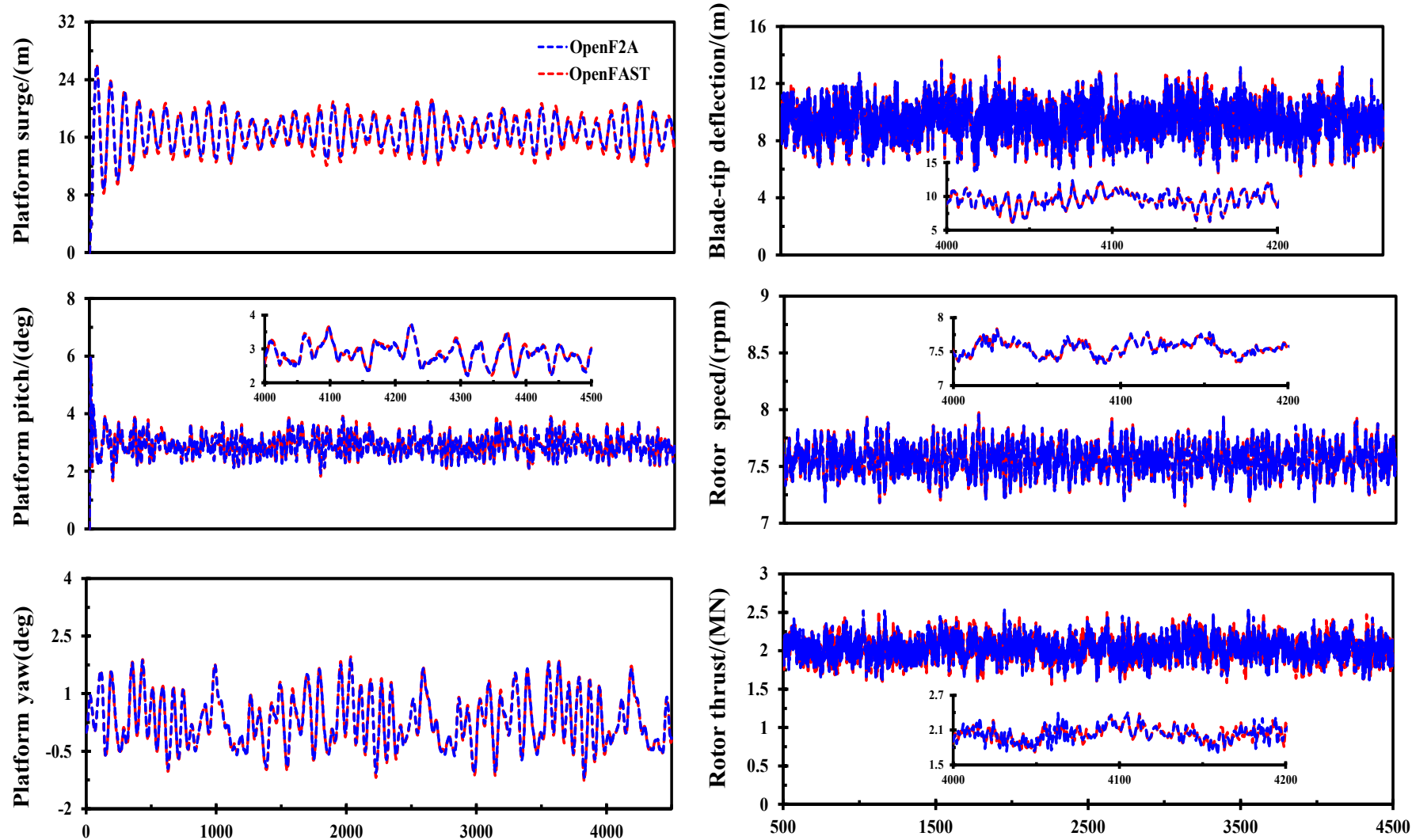
5MW
OC3Spar

10MW OOStar model

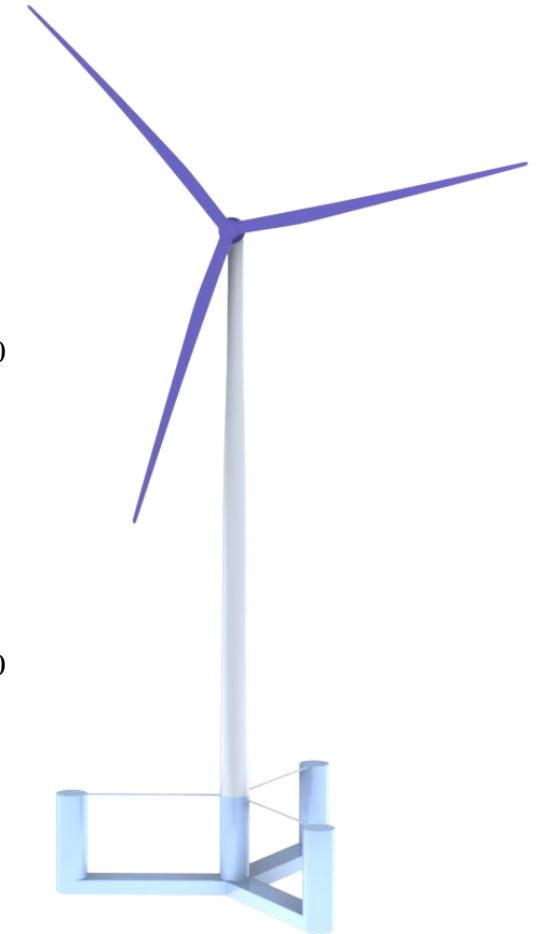
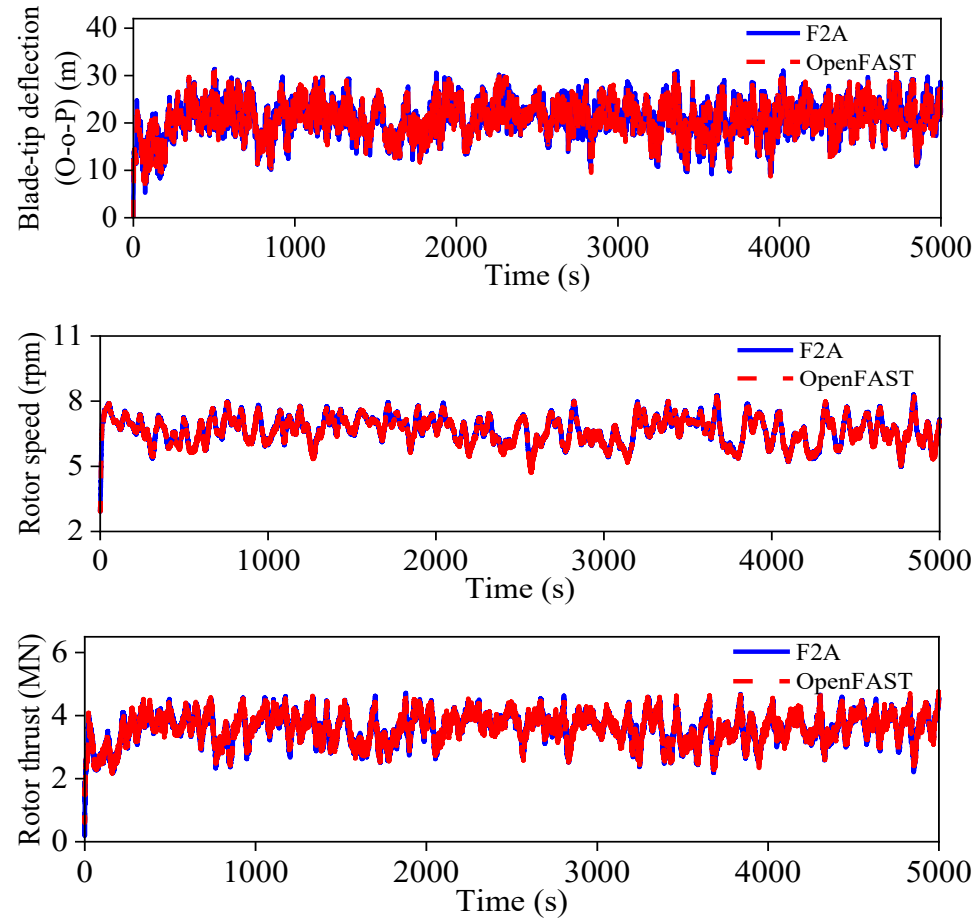
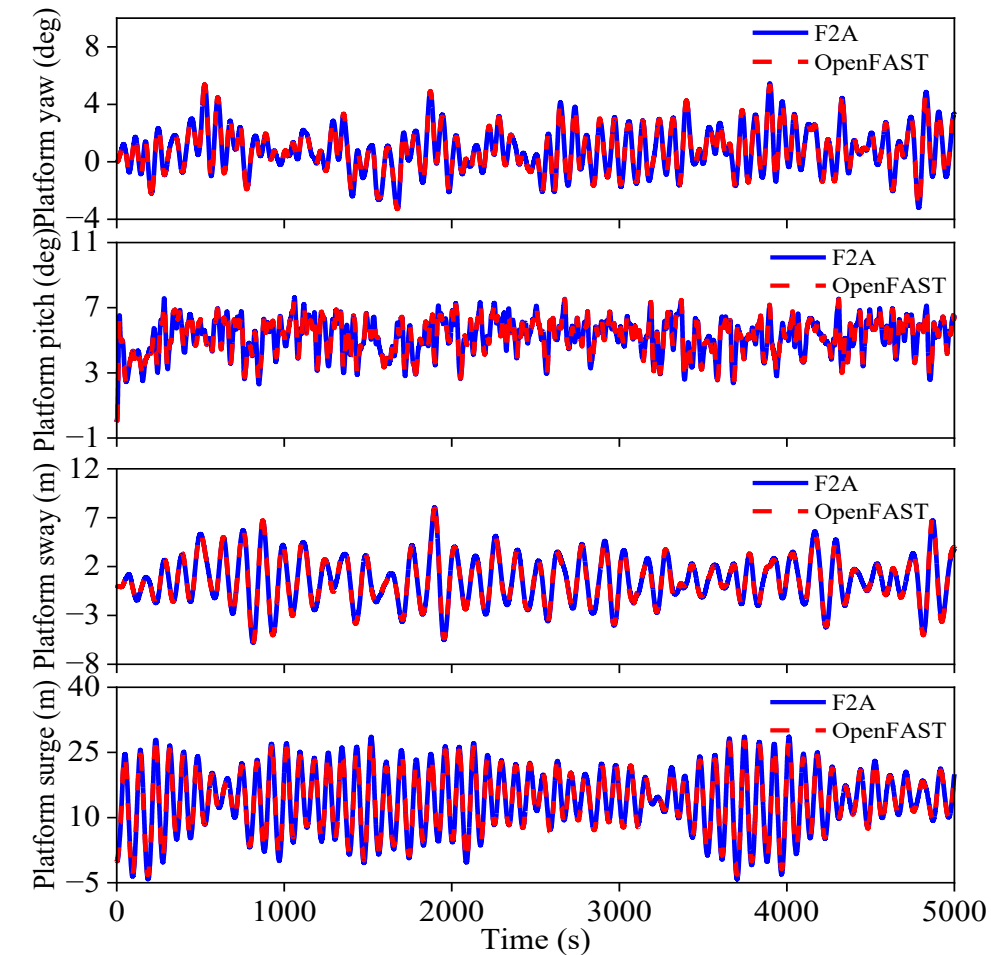


10MW
OOStar model

15MW VolturUS-S model



IEA22MW VolturUS-S model



IEA22MW
VolturUS-S model(scaled up
based on 15MW VolturUS)



User guide



4. User guide

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Open source files

OpenF2A is released on Github, which can accessed by: <https://github.com/yang7857854/OpenF2A>

 **OpenF2A** Public

Unpin Watch 0 Fork 2 Starred 7

main 1 Branch 1 Tag t Add file Code

 **yang7857854** Update README.md 1ed12b4 · 1 minute ago 34 Commits

 CADFiles	Update CAD files of the models	29 minutes ago
 DLL	OpenF2A v.4.0.0	1 hour ago
 RegTests	OpenF2A v.4.0.0	1 hour ago
 Release	OpenF2A v.4.0.0	1 hour ago
 SourceCode	OpenF2A v.4.0.0	1 hour ago
 LICENSE	Initial commit	4 months ago
 README.md	Update README.md	1 minute ago

About

Fully coupled framework for the dynamic analysis of floating wind energy systems (wind turbine, wind-wave hybrid concepts) based on OpenFAST and AQWA. The precompiled DLL and regression tests are provided.

-  Readme
-  Apache-2.0 license
-  Activity
-  7 stars
-  0 watching
-  2 forks



DLL files

OpenF2A is actually a DLL that is invoked by AQWA to calculate external loads on the floating bodies.

Release

https://github.com/yang7857854/OpenF2A/releases/tag/OpenF2A_v4.0.0

Assets 9

 InputforOpenF2A.txt	sha256:db8d4fff82d1451e24c98cf4e1750...		1.42 KB	34 minutes ago
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 readAQWAResults_SingleFile.exe	sha256:2c451809f327057aa37c28f80f273...		68 KB	34 minutes ago
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 Source code (zip)	Nov 18, 2025			
 Source code (tar.gz)	Nov 18, 2025			

Use OpenF2A

Case study of the 5MW DeepCWind model

Step1: Download essential files

▼ Assets 9

 [InputforOpenF2A.txt](#)

Example input of OpenF2A

 [InputforreadAQWAResults_SingleFile.txt](#)

Input for the result extraction code

 [openfast_x64_4.0.exe](#)

 [readAQWAResults_SingleFile.exe](#)

Extract results of AQWA

 [TurbSim_x64.exe](#)

TurbSim to generate wind data

 [user_force64-OpenF2A-4.0-option.dll](#)

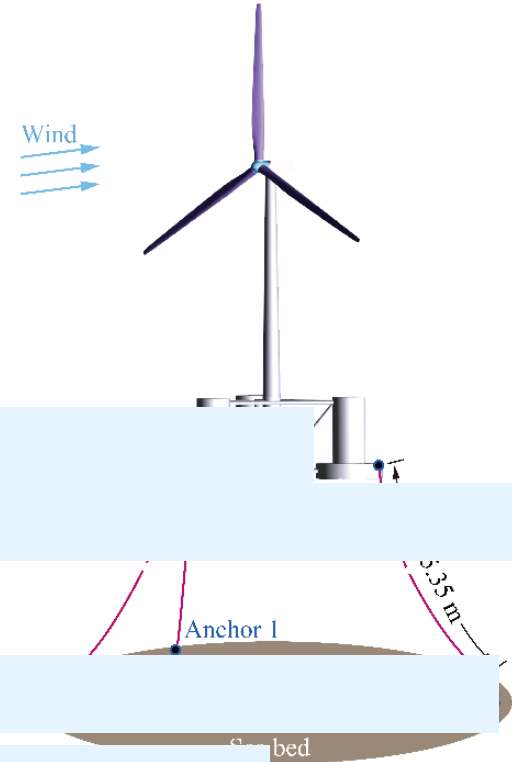
 [user_force64-OpenF2A-v4.0.0.dll](#)

DLL to be used

 [Source code \(zip\)](#)

Examples, source code and other files given in the main repository

 [Source code \(tar.gz\)](#)



Use OpenF2A

Step2: rename, copy and replace the userforce DLL

Rename it as “user_force64.dll”, copy it to the path of AQWA bin files, replace the DLL

The bin file path in my computer is: D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64

Program Files > ANSYS Inc > v241 > aqwa > bin > winx64

名称	修改日期	类型	大小
user_force64.dll	3/9/2026 12:44 AM	应用程序扩展	47,085 KB
AqwaWave.exe	11/6/2023 8:24 PM	应用程序	96,232 KB
Ags.exe	11/6/2023 8:24 PM	应用程序	147,691 KB
Aqwa.exe	11/6/2023 8:24 PM	应用程序	141,139 KB
Aqwa2Neut.exe	11/6/2023 8:24 PM	应用程序	84,804 KB
AqwaFlow.exe	11/6/2023 8:24 PM	应用程序	84,799 KB
wdrv.dll	11/6/2023 8:24 PM	应用程序扩展	80 KB
ApipWrapper.dll	11/6/2023 8:03 PM	应用程序扩展	957 KB
libcrypto-1_1-x64.dll	10/5/2023 7:38 PM	应用程序扩展	2,781 KB
libssl-1_1-x64.dll	10/5/2023 7:38 PM	应用程序扩展	669 KB
ag.tin	8/29/2023 1:48 PM	ANSYS 2020 R1	1,947 KB
zlib1.dll	8/2/2023 6:27 PM	应用程序扩展	130 KB



Use OpenF2A

Step3: open command prompt window, change path to AQWA bin files

- (1) Press <win> + <R>, type “cmd”, or click ‘start’, type “CMD” to open the command prompt window
- (2) Change directory to the AQWA bin folder using “cd” command

```
命令提示符
Microsoft Windows [版本 10.0.19045.6466]
(c) Microsoft Corporation。保留所有权利。

C:\Users\YY>D:

D:\>cd D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64

D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64>
```

On my computer, the AQWA bin path is located in the D drive. We need to change the drive index, and then change directory.

- a) Type D:
- b) Press Enter
- c) Type cd D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64
- d) Press Enter



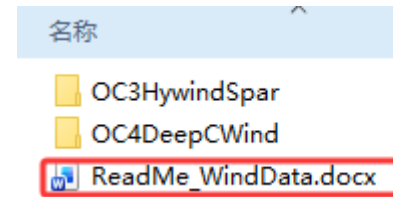
Use OpenF2A

Copy the TurbSim program downloaded previously to the path of wind data that is:

```
*/OpenF2A/RegTests/OC4DeepCWind/5MW_Baseline/  
Wind/.
```

Run TurbSim to generate the wind data, following the description in the document:
ReadMe_WindData.docx located in “RegTests”

名称	修改日期	类型	大小
NREL5MW_8.hh	10/24/2025 11:51 PM	C++ source file	4,670 KB
NREL5MW_8.inp	10/24/2025 11:49 PM	Notepad++ Doc...	8 KB
NREL5MW_11.4.hh	10/24/2025 11:52 PM	C++ source file	4,662 KB
NREL5MW_11.4.inp	10/24/2025 11:50 PM	Notepad++ Doc...	8 KB
NREL5MW_20.hh	10/24/2025 11:58 PM	C++ source file	4,653 KB
NREL5MW_20.inp	10/24/2025 11:50 PM	Notepad++ Doc...	8 KB
TurbSim_x64.exe	7/18/2022 1:41 PM	应用程序	26,355 KB



•Note•

Because of size limitations, the full field wind file for turbulent wind conditions cannot be uploaded on GitHub. Therefore, you may encounter a problem when trying to manage the simulation. Therefore, please run TurbSim to generate the wind field data based on the input in directory "SMW_BaseLine\Wind\" by simply dragging an ".inp" file on TurbSim.exe, or following the syntax in a CMD promote window below:

1. Launch cmd prompt window
<Win>+R, or click "Start", then type "cmd" and press <Enter>



2. Change drive
Type your drive ID followed by ":", like F: as follows:



3. change directory to the TurbSim's path using the command of "cd", like:



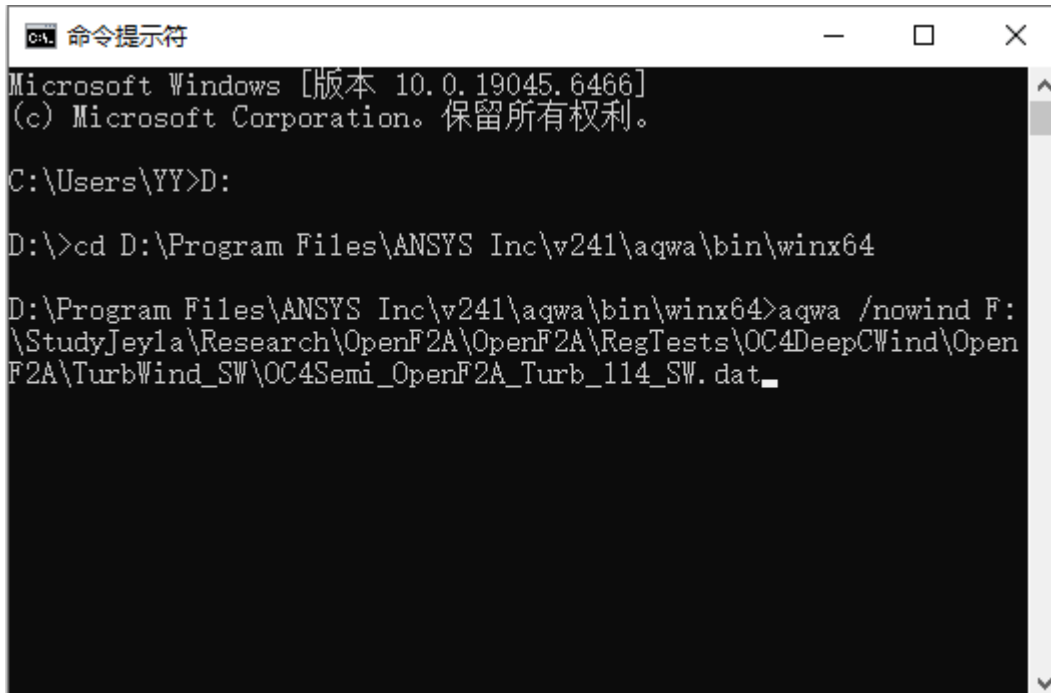
4. Run TurbSim

Type executable file name followed by a space and full name of TurbSim input file, like:
TurbSim_x64.exe NREL5MW_11.4.inp

Use OpenF2A

Step5: Run AQWA program

Run AQWA program following the syntax: `aqwa /nowind <full path of AQWA case input file>`



```
命令提示符
Microsoft Windows [版本 10.0.19045.6466]
(c) Microsoft Corporation。保留所有权利。

C:\Users\YY>D:

D:\>cd D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64

D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64>aqwa /nowind F:\StudyJeyla\Research\OpenF2A\OpenF2A\RegTests\OC4DeepCWind\OpenF2A\TurbWind_SW\OC4Semi_OpenF2A_Turb_114_SW.dat_
```

`aqwa`: denotes the program name of AQWA

`/nowind`: denotes not to displace the progress window of AQWA case

`<full path of AQWA case input file>`: denotes the full path of the AQWA case input file.

Press <Enter> to run the program

The full path of the case file on my computer is:
F:\StudyJeyla\Research\OpenF2A\OpenF2A\RegTests\OC4DeepCWind\OpenF2A\TurbWind_SW\OC4Semi_OpenF2A_Turb_114_SW.dat

Use OpenF2A

Step6: Generate results

The notice of OpenF2A and OpenFAST subroutine output information will be displayed. The results stored in the <*.out> and <*.lis> files at anytime you want.

```
D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64\aqwa /nowind F:\StudyJeyla\Research\OpenF2A\OpenF2A\Re
gTests\OC4DeepCWind\OpenF2A\TurbWind_SW\OC4Semi_OpenF2A_Turb_114_SW.dat

-----
OpenF2A
(v4.00.00-yy, 24-June-2025)
-----

OpenF2A is originally developed by Professor Yang Yang in Ningbo University
by incorporating OpenFAST with AQWA for conducting fully-coupled simulations
of floating offshore wind turbines. The source code is released to the public
for free, which can be found on Github repository of yang7857584/OpenF2A
-----

There is 1 wind turbine considered.
Using FAST input file: 5MW_OC4Semi_F2A_Turb_11.4.fst

*****
OpenFAST

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Copyright (C) 2025 Envision Energy USA LTD

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See the "LICENSE" file distributed with this software for details.
*****

OpenFAST-unversioned
Compile Info:
- Compiler: Intel(R) Fortran Compiler 2021
- Architecture: 64 bit
- Precision: single
- OpenMP: No
```

名称	修改日期
5MW_OC4Semi_F2A_Turb_11.4.fst	10/24/2025 11:57 PM
5MW_OC4Semi_F2A_Turb_11.4.out	3/9/2026 7:08 PM
InputforOpenF2A.txt	3/9/2026 10:04 AM
OC4Semi_OpenF2A_Turb_114_SW.dat	11/9/2025 2:15 PM
OC4SEMI_OPENF2A_TURB_114_SW.DCP	3/9/2026 7:05 PM
OC4SEMI_OPENF2A_TURB_114_SW.LIS	3/9/2026 7:08 PM
OC4SEMI_OPENF2A_TURB_114_SW.MES	3/9/2026 7:05 PM
OC4SEMI_OPENF2A_TURB_114_SW.PLD	3/9/2026 7:05 PM
OC4SEMI_OPENF2A_TURB_114_SW.POS	3/9/2026 7:05 PM
OC4SEMI_OPENF2A_TURB_114_SW.RES	3/9/2026 7:08 PM

```
Time: 0 of 3600 seconds.
The BEM solution is being turned off due to low TSR. (TSR = 0). This warning will not be
repeated though the condition may persist. (See GeomPhi output channel.)

FAST_Solution:FAST_UpdateStates:CalcOutputs_And_SolveForInputs:SolveOption2:AD_CalcOutput:RotCalcO
utput:BEMT_CalcOutput(node 5, blade 1):UA_CalcOutput:UA_BlendSteady:Temporarily turning off UA
due to high angle of attack or low relative velocity. This warning will not be repeated though
the condition may persist.
Time: 9 of 3600 seconds. Estimated final completion at 19:42:46.
```



The *.out file is generated by OpenFAST. The *.lis is generated by AQWA, in which the results can be extracted using “readAQWAResults_SingleFile.exe” downloaded previously.

```

1 *****1*****2*****3*****4*****5*****6*****7*****8
2 23456789012345678901234567890123456789012345678901234567890
3 *****
4 ***** File generated by Aqwa in Workbench 2024 R1 *****
5 *****
6 * Project Title :
7 * Project Reference :
8 * Project Author :
9 * Project Description :
10 * Date of Creation : 2025/2/19 16:32:56
11 * Last Modified : 2025/2/19 16:51:36
12 * This analysis file written at : 2025/2/19 18:24:44
13 * Hydrodynamic Solver Unit System : Metric: kg, m [N]
14 *****
15 ***** DECK 0 *****
16 *****
17
18 DATE:09/03/26 TIME:19:08:27
19
20 JOB AQWA DRIF WFRQ
21
22 TITLE
23
24 NUM_CORES 3
25
26 OPTIONS CONV FDLL
27
28 OPTIONS NOST
29
30 OPTIONS REST END
31
32 RESTART 1 5
33
34
35 Aqwa-Drift - 2024 R1 64BIT VERSION 2024 R1
36 License Name: aqwa_solve
37 License Type: Commercial
38
39
40 Point Releases and Patches installed:
41
42 Ansys, Inc. License Manager 2024 R1
43 Viewer 2024 R1
44 Aqwa 2024 R1
45 SpaceClaim 2024 R1
46 Customization Files for User Programmable Features 2024 R1
47 Mechanical Products 2024 R1

```

Use OpenF2A

Step8: Extract results from the LIS file

Modify the “inputforreadAQWAResults_SingleBody.txt” properly, then double click “readAQWAResults_SingleFile.exe”.

```

1  ! Input file for readAQWAResults_SingleBody which is specific developed for extracting results from the LIS files generated by AQWA.
2  ! The positions, velocities and accelerations of the platform can be output based on the definitions below. The platform is a single
3  ! ----- Global parameters -----
4  "OC4SEMI_OPENF2A_TURB_114_SW.LIS"          - InputLISFile          - Input LIS file
5  "OC4SEMI_OPENF2A_TURB_114_SW"              - OutFolder              - The target folder to store the result files
6  3600.0                                     - TimeDur               - Simulation length
7  0.1                                        - TimeStep              - Time step
8  ! ----- Details of the LIS -----
9  1                                           - NumBodies             - Number of bodies defined in the LIS file.
10 !----- Body 1 -----
11 "Semisub"                                - NameBodies(1)         - Name of the 1st body
12 True                                     - FlagOPMot(1)          - Flag of outputting motions at the 1st body's CoG
13 True                                     - FlagOPVel(1)          - Flag of outputting velocities at the 1st body's CoG
14 True                                     - FlagOPAcc(1)          - Flag of outputting accelerations at the 1st body's CoG
15 True                                     - FlagOPFrc(1)          - Flag of outputting forces at the 1st body's CoG
16 !----- Added points -----
17 0                                           - NumAddedPts           - Number of added points that require output results (accelerations only)
18 ! ----- Tension of the mooring lines -----
19 True                                     - FlagMLType            - Mooring line type (T: Tension, F: Force, A: Acceleration, M: Motion, V: Velocity)
20 3                                           - NumMoorLines          - Number of mooring lines
21 ! ----- Wave elevation -----
22 True                                     - FlagWvElev            - Output wave elevation
23 "WAVE PSN NODE 99999-FIXED" - CharWvEle            - Characteristic string of the wave elevation line.

```

Simulation length and time step defined in AQWA

Mooring lines

Wave kinematics at [0,0,0] are extracted by default

Use OpenF2A

Step8: Extract results from the LIS file

Extraction of result will start as follows

选择 F:\StudyJeyla\Research\1Papers\2025English\OpenF2A_Case4.0\O

```
InputLISFile= OC4SEMI_OPENF2A_TURB_114_SW.LIS
Time = 0 s
Time = 100 s
Time = 200 s
Time = 300 s
Time = 400 s
Time = 500 s
Time = 600 s
Time = 700 s
Time = 800 s
Time = 900 s
Time = 1000 s
Time = 1100 s
Time = 1200 s
Time = 1300 s
Time = 1400 s
Time = 1500 s
Time = 1600 s
Time = 1700 s
Time = 1800 s
Time = 1900 s
Time = 2000 s
Time = 2100 s
Time = 2200 s
Time = 2300 s
```

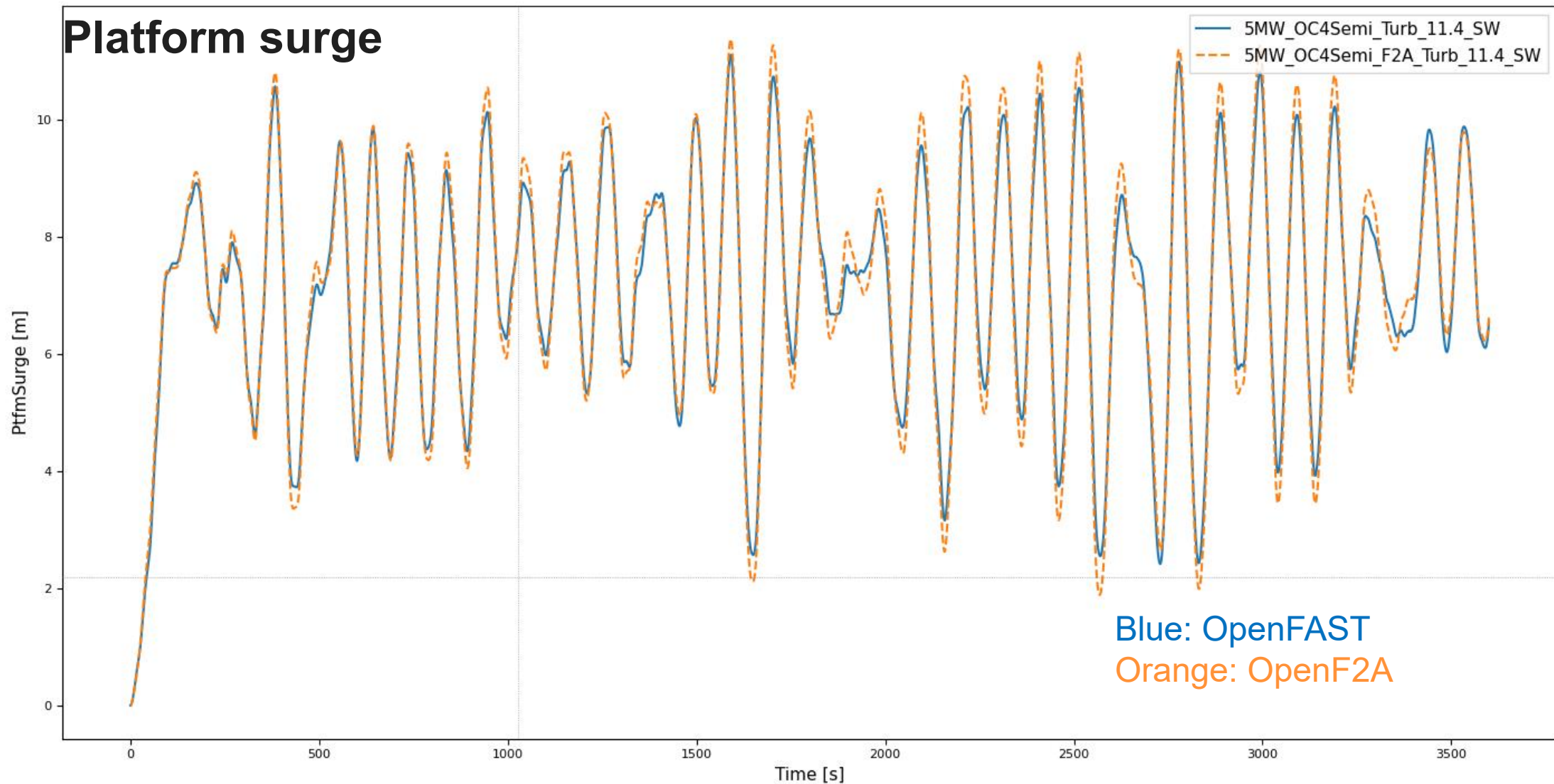
Data in the LIS file are extracted and written in the files

OC4SEMI_OPENF2A_TURB_114_SW

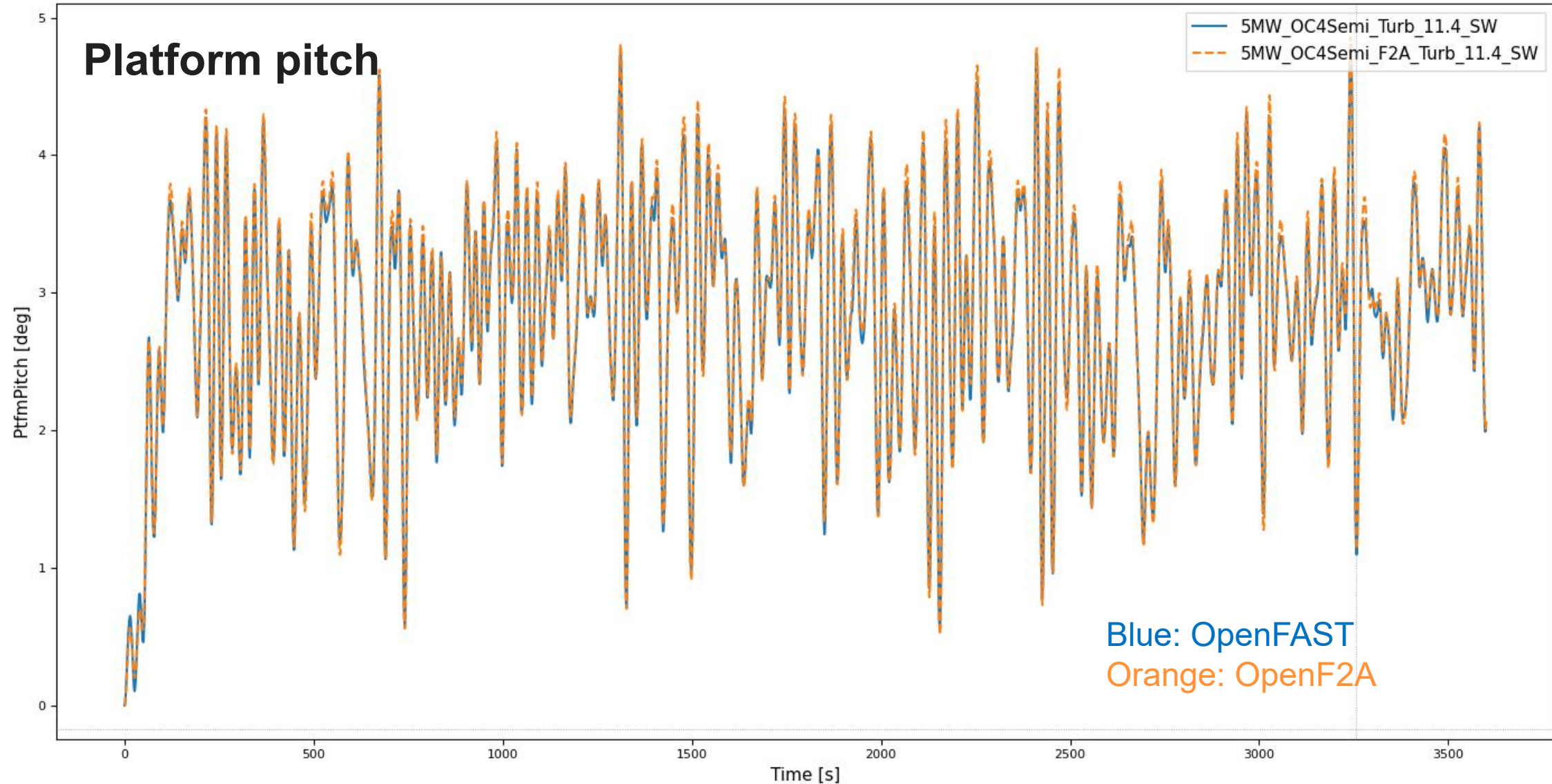
- OC4SEMI_OPENF2A_TURB_114_SW_MooringTension.dat 3/9/2026 7:19 PM
- OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Acceleration.dat 3/9/2026 7:19 PM
- OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Force.dat 3/9/2026 7:19 PM
- OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Motion.dat 3/9/2026 7:19 PM
- OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Velocity.dat 3/9/2026 7:19 PM
- OC4SEMI_OPENF2A_TURB_114_SW_WaveAcceleration.dat 3/9/2026 7:19 PM
- OC4SEMI_OPENF2A_TURB_114_SW_WaveElevation.dat 3/9/2026 7:19 PM
- OC4SEMI_OPENF2A_TURB_114_SW_WaveVelocity.dat 3/9/2026 7:19 PM

Tension of mooring lines of the OC4SEMI_OPENF2A_TURB_114_SW\OC4SEMI_OPENF2A_TURB_114_SW case extracted							
	Time	Mooring1_FL	Mooring2_FL	Mooring3_FL	Mooring1_AC	Mooring2_AC	Mooring3_AC
	s	(N)	(N)	(N)	(N)	(N)	(N)
4	0.000	1.09640E+06	1.09640E+06	1.09640E+06	8.99110E+05	8.99110E+05	8.99110E+05
5	0.100	1.09550E+06	1.09570E+06	1.09540E+06	8.99110E+05	8.99110E+05	8.99110E+05
6	0.200	1.09450E+06	1.09490E+06	1.09430E+06	8.99100E+05	8.99110E+05	8.99100E+05
7	0.300	1.09350E+06	1.09410E+06	1.09330E+06	8.99020E+05	8.99030E+05	8.99000E+05
8	0.400	1.09250E+06	1.09350E+06	1.09220E+06	8.98120E+05	8.98280E+05	8.98010E+05
9	0.500	1.09170E+06	1.09300E+06	1.09130E+06	8.96130E+05	8.96770E+05	8.95890E+05
10	0.600	1.09090E+06	1.09240E+06	1.09050E+06	8.94440E+05	8.95430E+05	8.94080E+05
11	0.700	1.08980E+06	1.09170E+06	1.08930E+06	8.92820E+05	8.94360E+05	8.92320E+05
12	0.800	1.08790E+06	1.09010E+06	1.08730E+06	8.91420E+05	8.93610E+05	8.90840E+05
13	0.900	1.08610E+06	1.08880E+06	1.08540E+06	8.90350E+05	8.92830E+05	8.89750E+05
14	1.000	1.08480E+06	1.08790E+06	1.08390E+06	8.89330E+05	8.92220E+05	8.88680E+05
15	1.100	1.08370E+06	1.08750E+06	1.08280E+06	8.87980E+05	8.91100E+05	8.87230E+05
16	1.200	1.08300E+06	1.08750E+06	1.08210E+06	8.86520E+05	8.89930E+05	8.85690E+05
17	1.300	1.08260E+06	1.08750E+06	1.08160E+06	8.85670E+05	8.89510E+05	8.84770E+05
18	1.400	1.08200E+06	1.08750E+06	1.08090E+06	8.85320E+05	8.89520E+05	8.84330E+05
19	1.500	1.08120E+06	1.08710E+06	1.08010E+06	8.85220E+05	8.90180E+05	8.84230E+05
20	1.600	1.08100E+06	1.08710E+06	1.07980E+06	8.85450E+05	8.90930E+05	8.84420E+05
21	1.700	1.08150E+06	1.08770E+06	1.08020E+06	8.85590E+05	8.91670E+05	8.84520E+05
22	1.800	1.08230E+06	1.08850E+06	1.08100E+06	8.85540E+05	8.92180E+05	8.84380E+05

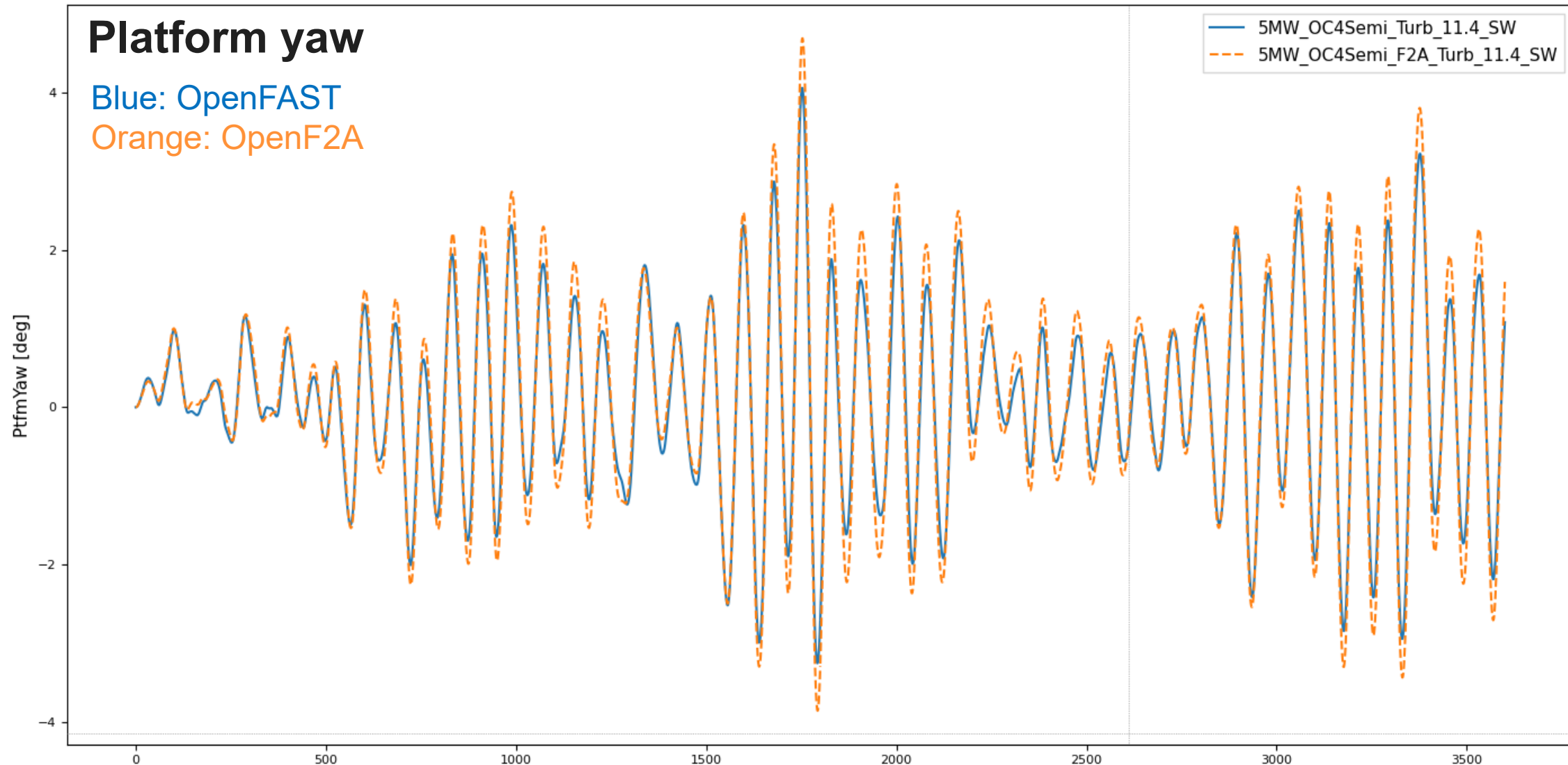
Comparison between OpenFAST and OpenF2A



Comparison between OpenFAST and OpenF2A



Comparison between OpenFAST and OpenF2A



References

The following publications are for your information when you need a reference to cite if OpenF2A is useful in your research. The 1st and 8th publications listed below are recommended the most. We are grateful for your acknowledgement by citing our research. More suitable references can be found on my google scholar website <https://scholar.google.com/citations?user=frTAwNQAAAAJ&hl=zh-CN>.

1. Yang, Y., Yu, J., Bashir, M., & Li, C. (2026). [Development and applicability of a novel fully coupled framework for floating wind energy systems](#). *Ocean Engineering*, 354, 124900
2. Hu, G., Ding, J., Lai, Y., He, C., Zhang, Z., Liu, W., ... & Yang, Y. (2026). Coupled dynamic behavior of a floating offshore wind turbine integrated with flapping-type-wave energy converters under wind-wave misalignment conditions. *Ocean Engineering*, 353, 124816.
3. Liu, W., ZHANG, Z., Lai, Y., Hu, G., Yu, J., Li, C., ... & Yang, Y. (2026). Stability and power performance of a floating wind-current energy system under tidal turbine operating mode variations. *Energy*, 140209.
4. Nie, D., Li, S., Yan, Y., Lou, Y., Yin, J., Hao, J., ... & Yang, Y. (2025). Structural analysis of a 15 MW floating offshore wind turbine platform based on a novel fully-coupled framework. *Ocean Engineering*, 342, 123057.
5. Yin, J., Fan, Y., Bashir, M., Nie, D., Lai, Y., Ding, J., ... & Yang, Y. (2025). Development of a hybrid deep learning model with HHO algorithm for dynamic response prediction of wind-wave integrated floating energy systems. *Ocean Engineering*, 340, 122394.
6. Ding, J., Yang, Y., Yu, J., Bashir, M., Ma, L., Li, C., & Li, S. (2024). Fully coupled dynamic responses of barge-type integrated floating wind-wave energy systems with different WEC layouts. *Ocean Engineering*, 313, 119453.
7. Yang, Y., Shi, Z., Fu, J., Ma, L., Yu, J., Fang, F., ... & Yang, W. (2023). Effects of tidal turbine number on the performance of a 10 MW-class semi-submersible integrated floating wind-current system. *Energy*, 285, 128789.
8. Yang, Y., Bashir, M., Michailides, C., Li, C., & Wang, J. (2020). [Development and application of an aero-hydro-servo-elastic coupling framework for analysis of floating offshore wind turbines](#). *Renewable Energy*, 161, 606-625.

Thank you!

